

The importance of genomic population diversity: from cancer etiology to translation

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Cancer Genomic Epidemiology Course 2026

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Outline

1

- Introduction to genetic epidemiology

2

- What is population diversity and why does it matter?
 - Population genetics and public health perspective

3

- Leveraging diverse populations to study cancer susceptibility
 - Genome-wide association studies and more



Introduction

Genetic Epidemiology

What is **genetic** epidemiology?

The study of how **heritable (germline genetic) factors** contribute to health and disease in *populations* and how genes interplay with environmental factors



GENETICS

This is how it works

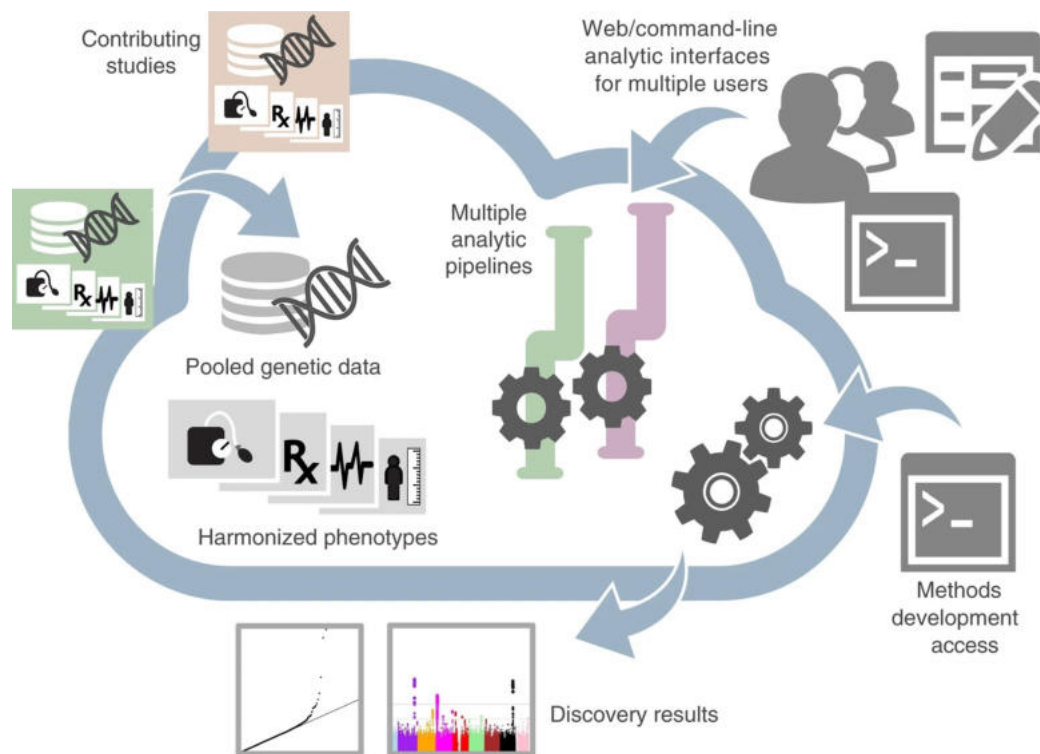


Introduction

Genetic Epidemiology

What is **genetic** epidemiology?

The study of how **heritable (germline genetic) factors** contribute to health and disease in *populations* and how genes interplay with environmental factors



Use theory and methods from **biostatistics, data science, bioinformatics, population genetics** to analyze data and interpret associations

Brody et al. *Nature Genetics* (2017) 49: 1560–1563

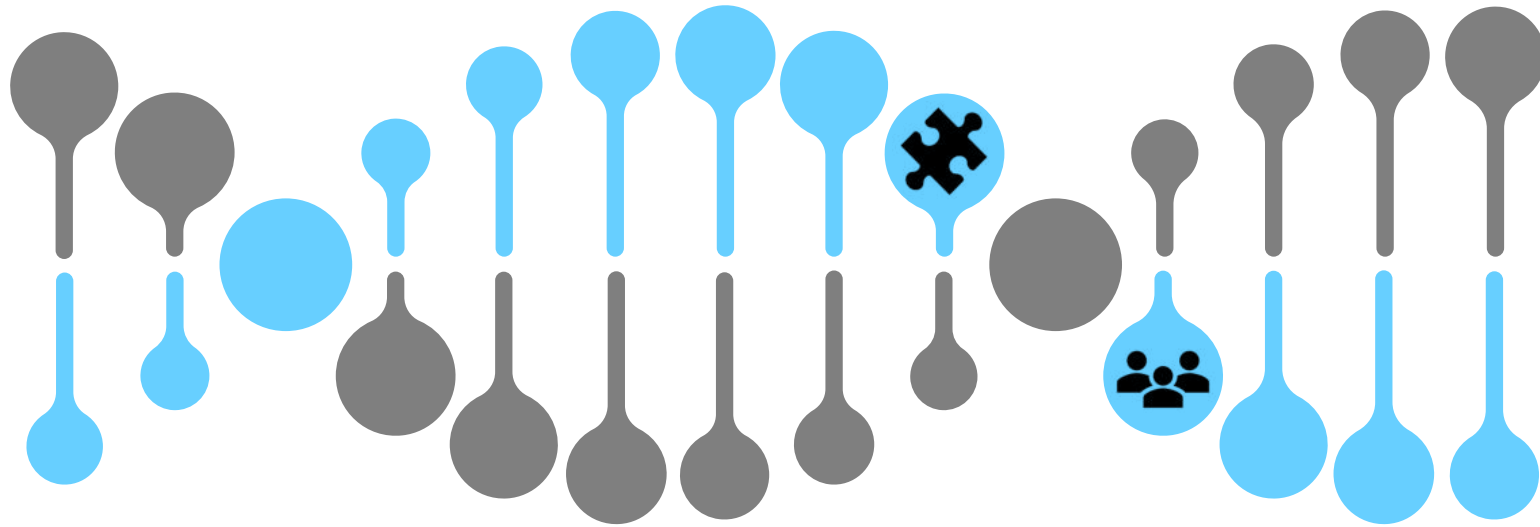


Introduction

Genetic Epidemiology

Why is **genetic** epidemiology useful?

Insight into Biology
Understand the causes
and mechanisms of
cancer susceptibility



Public Health

Identify people with high
risk of developing cancer

Clinical Applications

Develop and personalize
therapeutic interventions

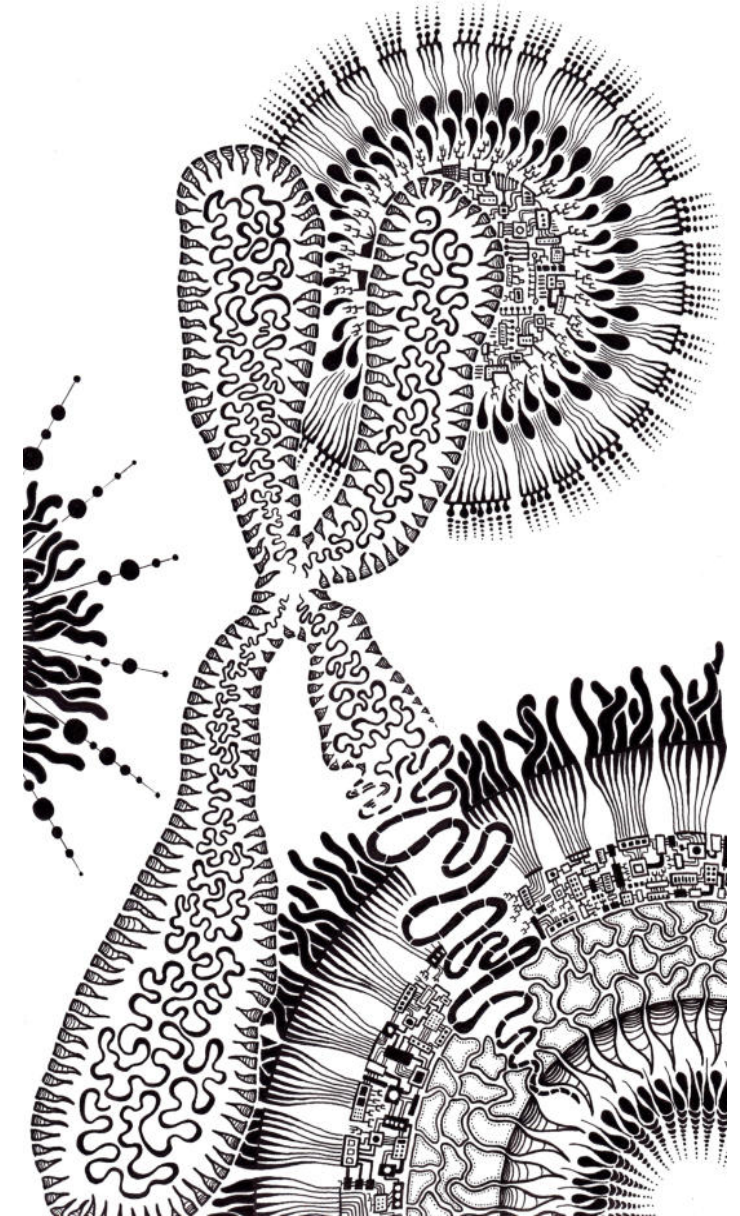


Introduction

Genetic Basis of Cancer

Cancer is a complex and multifactorial disease

- Substantial contribution of germline genetics to etiology beyond familial cancer predisposition syndromes
 - Heritability ranges from 7% to 58%



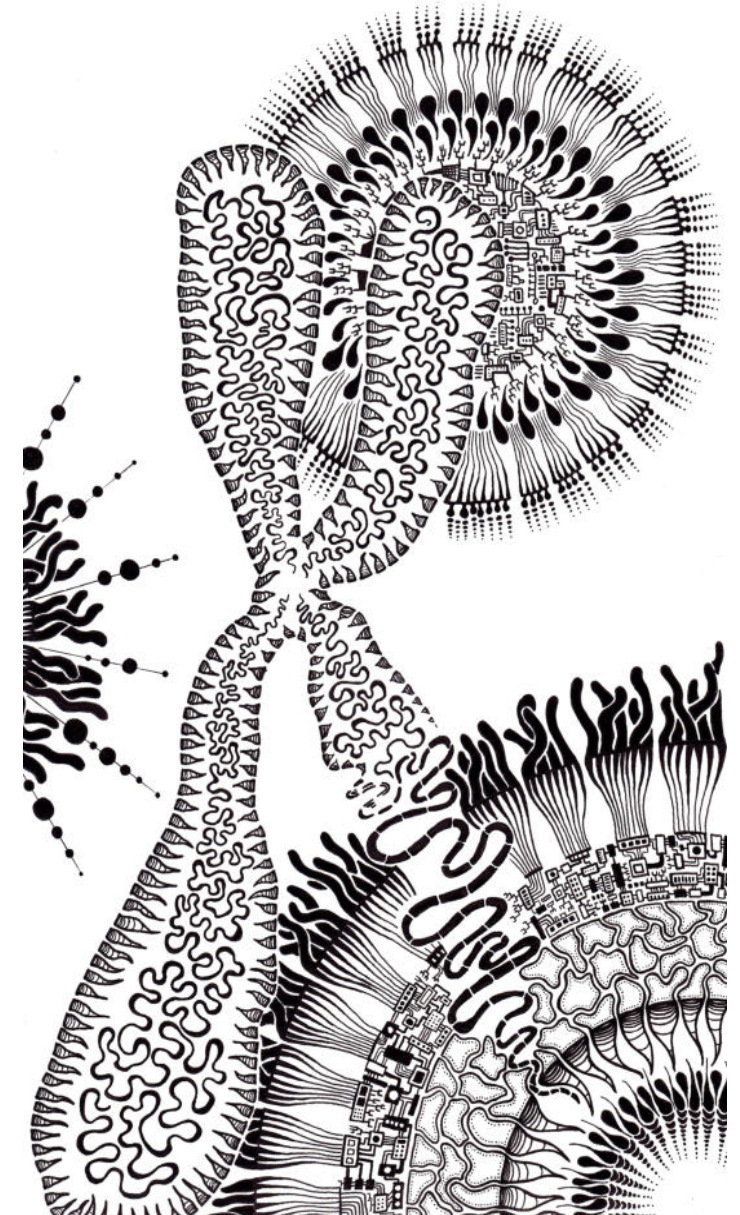


Introduction

Genetic Basis of Cancer

Cancer is a complex and multifactorial disease

- Substantial contribution of germline genetics to etiology beyond familial cancer predisposition syndromes
 - Heritability ranges from 7% to 58%
- Features of the germline genetic landscape:
 - No single gene is sufficient to cause cancer
 - Genetic variants at many different loci have modest effects on risk
 - Cancer susceptibility variants often influence multiple cancers and non-cancer phenotypes (**pleiotropy**)



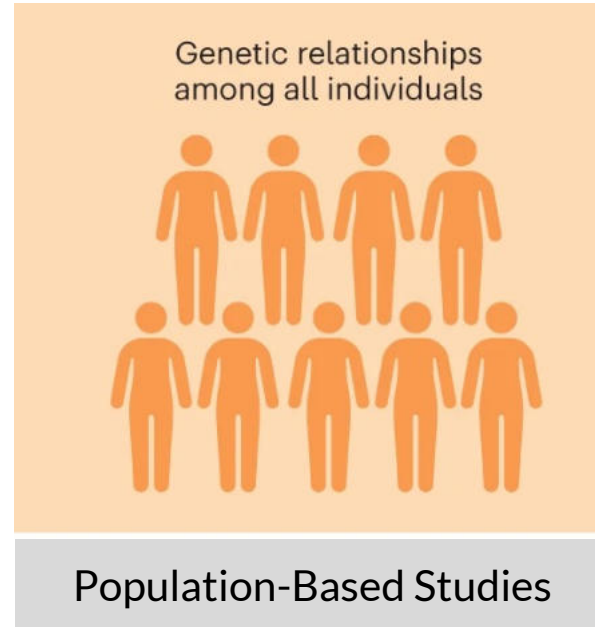
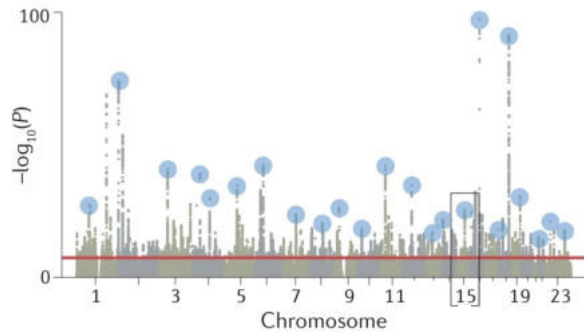


Introduction

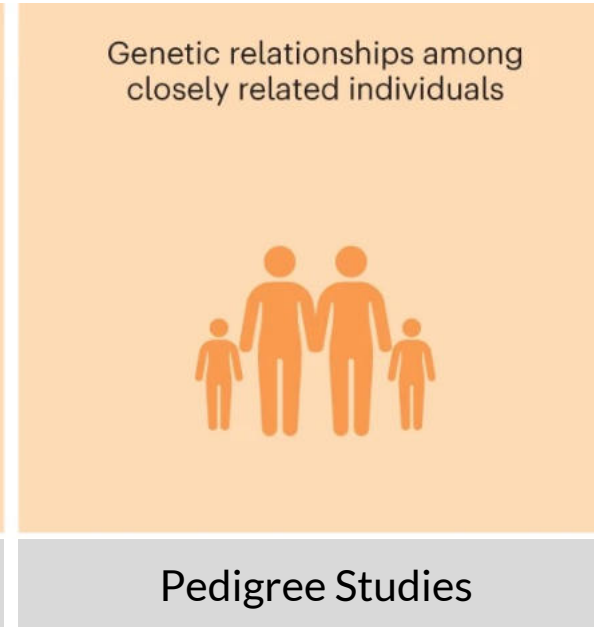
Genetic Basis of Cancer

Association vs Linkage

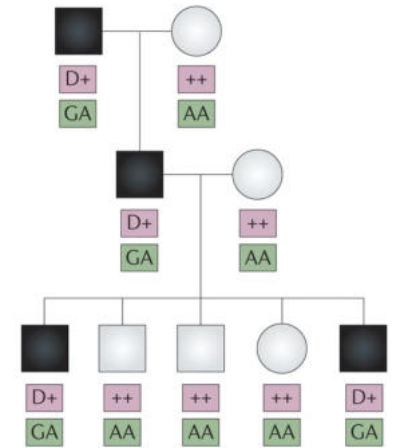
Genome-wide association studies (GWAS)



Compare allele frequencies between unrelated individuals



Segregation of loci and traits within families





Introduction

GWAS

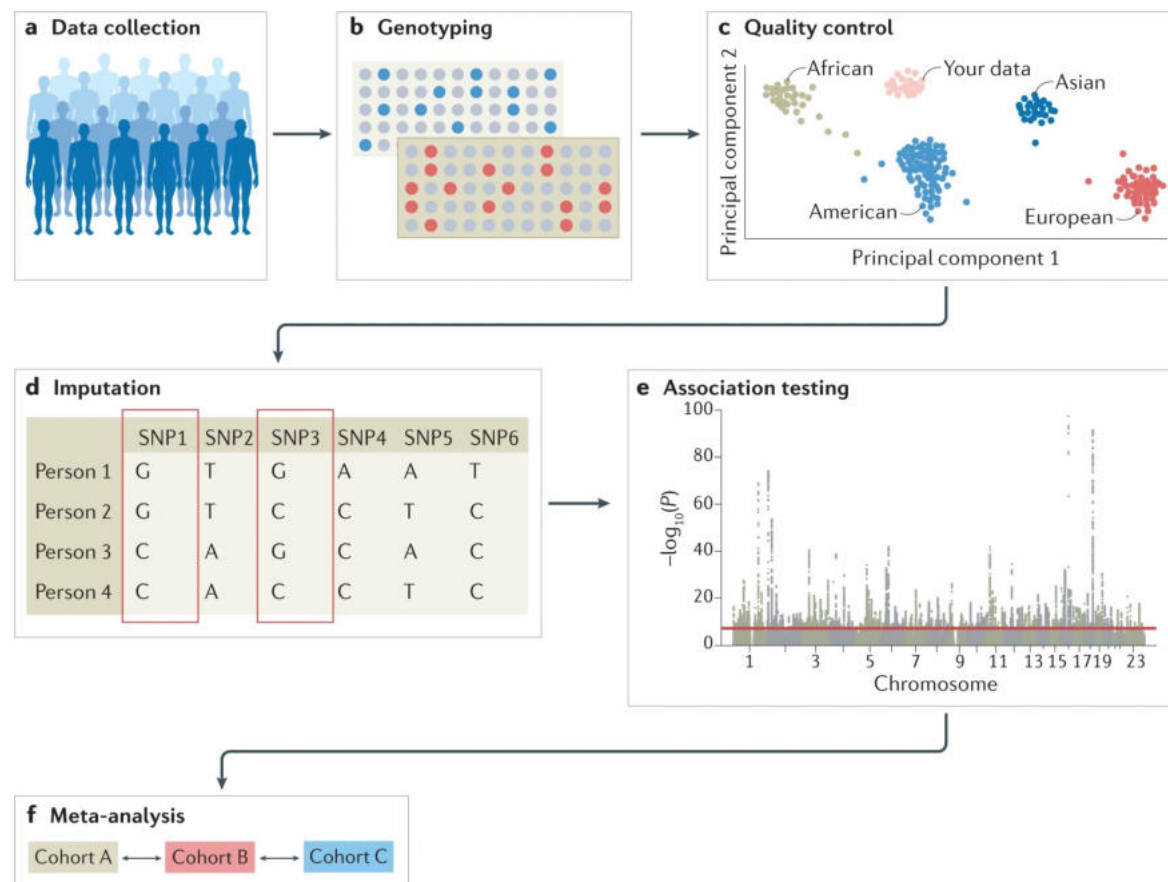
Genome-wide association studies (GWAS)

- Case-control studies of unrelated individuals
 - Mostly sporadic cancer cases
- Hypothesis-free scan of millions of common ($MAF \geq 0.01$) variants across the genome
 - Associations with **p-value $< 5 \times 10^{-8}$** are considered statistically significant, lower p-values do not imply causality

Genome-wide association studies

[Emil Uffelmann](#), [Qin Qin Huang](#), [Nchangwi Syntia Munung](#), [Jantina de Vries](#), [Yukinori Okada](#), [Alicia R. Martin](#), [Hilary C. Martin](#), [Tuuli Lappalainen](#) & [Danielle Posthuma](#) ✉

[Nature Reviews Methods Primers](#) **1**, Article number: 59 (2021) | [Cite this article](#)





Part I:

What is genomic population diversity?

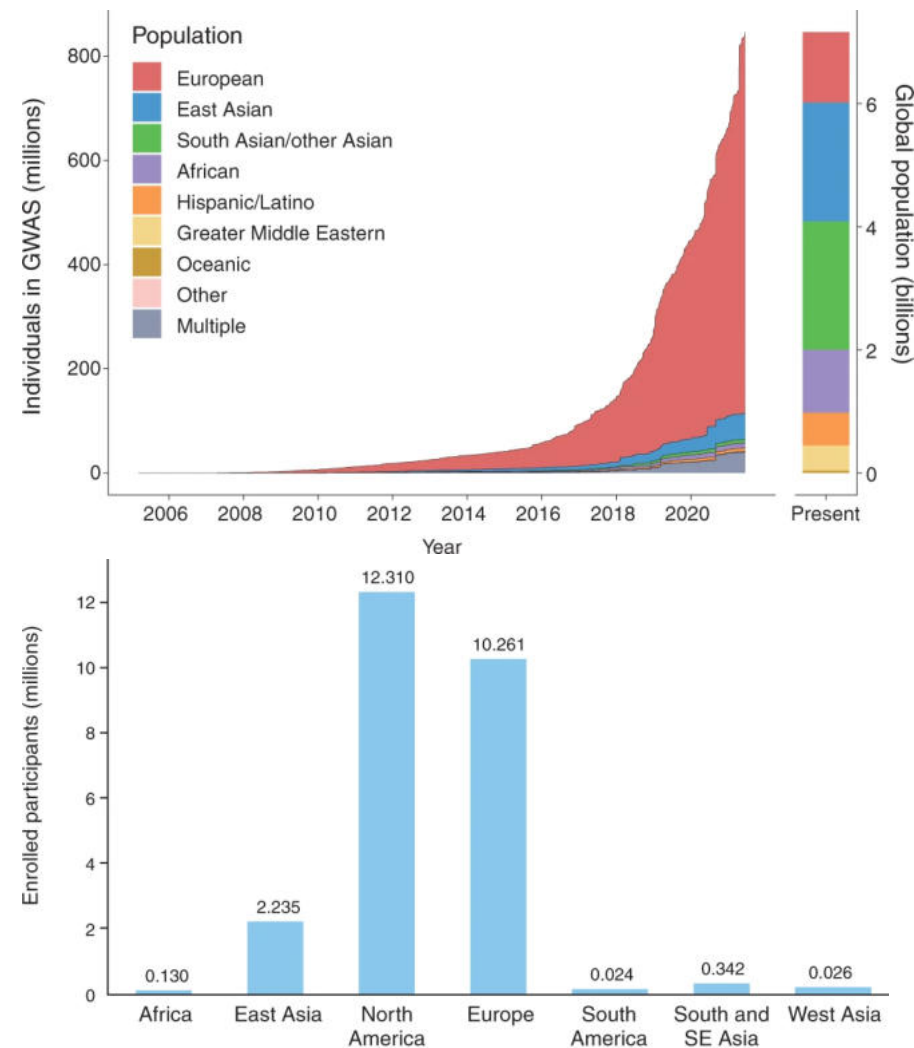
Evolutionary and public health perspective



Motivation

Why is population diversity important?

- Disparity in representation exacerbates health disparities
 - Under-representation in GWAS propagates to all aspects of genomic research
- Eurocentric bias limits the translational impact of genomic discoveries





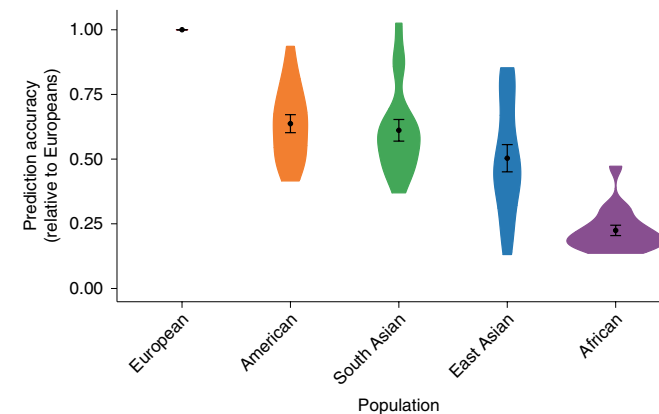
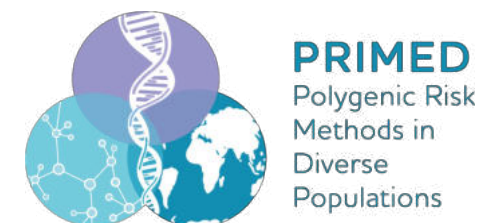
Motivation

Why is population diversity important?

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 - Genetic risk assessment using polygenic risk scores (PRS)

Principles and methods for transferring polygenic risk scores across global populations

Linda Kachuri^{1,2}, Nilanjan Chatterjee³, Jibril Hirbo^{4,5}, Daniel J. Schaid⁶, Iman Martin⁷, Iftikhar J. Kullo⁸, Eimear E. Kenny^{9,10,11}, Bogdan Pasaniuc^{12,13,14}, Polygenic Risk Methods in Diverse Populations (PRIMED) Consortium Methods Working Group*, John S. Witte^{1,2,15,16} & Tian Ge^{17,18,19}✉



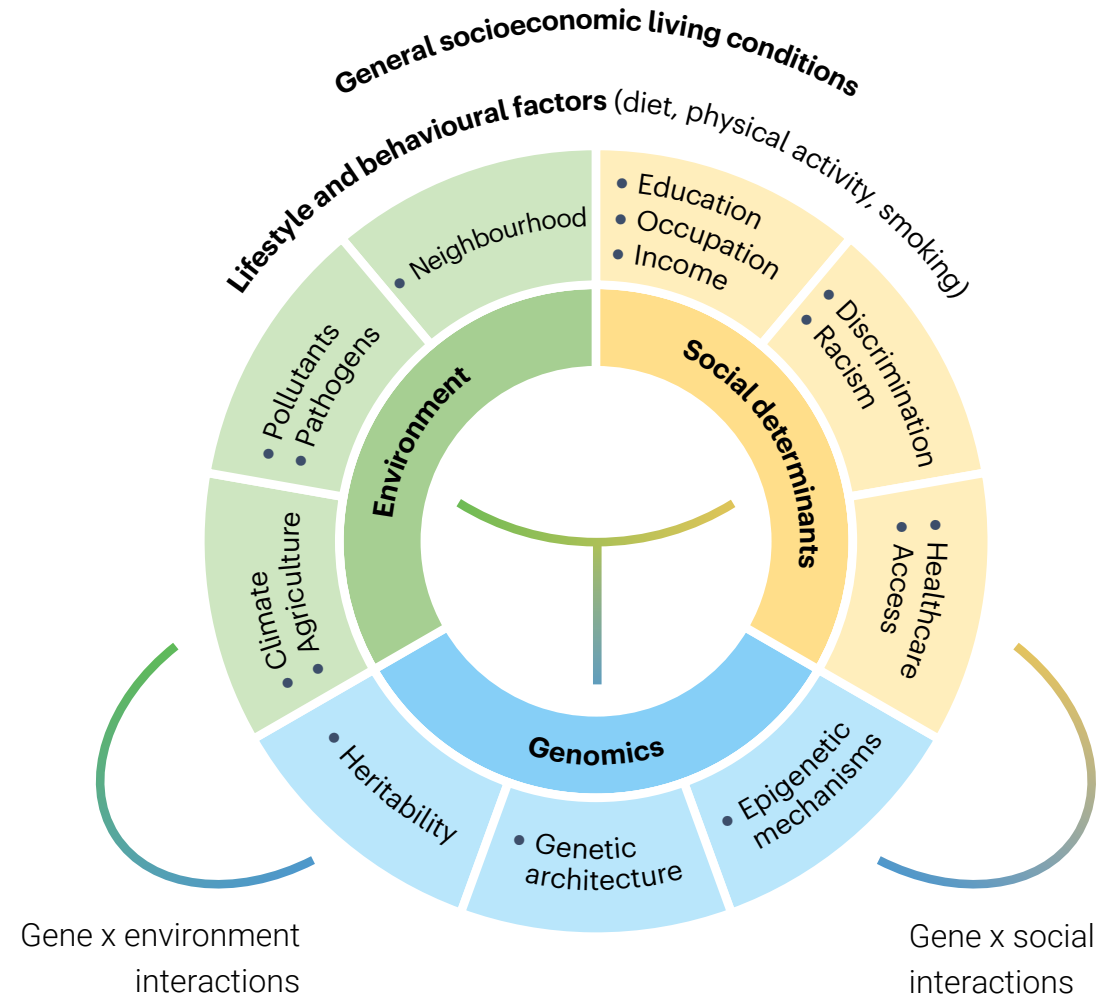


Motivation

Why is population diversity important?

More complete characterization of the mechanisms of cancer susceptibility

- Genetic architecture
 - Evolutionary and demographic forces
- Interplay between risk factors
 - Greater variability in genetic, social, and environmental contexts

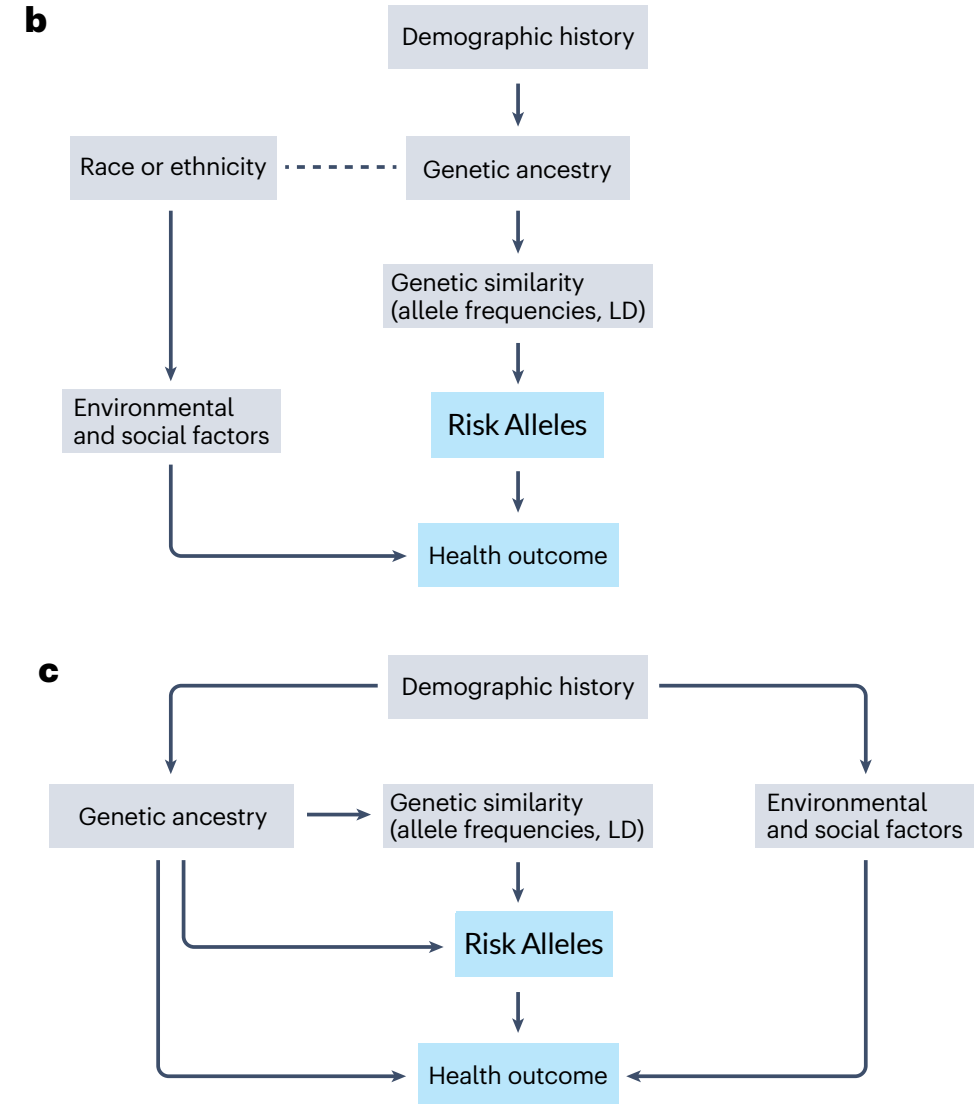




Motivation

Considerations for studies of diverse populations

- Genetic ancestry inference
- How do we define populations?
 - Socio-cultural groups
 - Clusters based on genetic similarity
- How do we assess population differences in genetic architecture?
 - Allele frequencies
 - Linkage disequilibrium (LD)



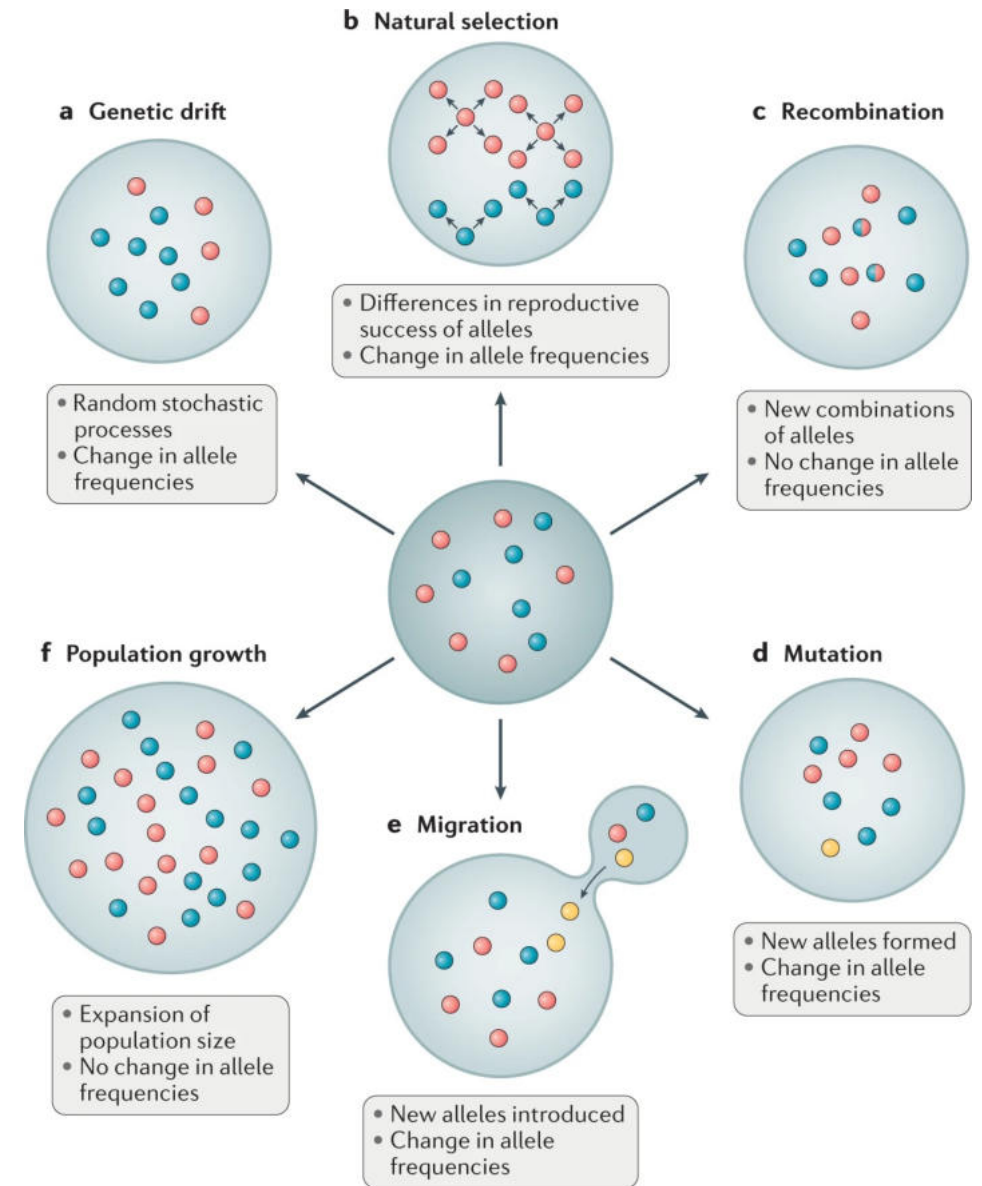


Key Concepts

Population Genetics

Our genomes contain a historical record of human evolution

- **New alleles** are introduced through **mutation** and **migration**
- Breakdown and **recombination** of DNA segments during meiosis changes the **LD structure** and creates new haplotypes
- **Genetic drift** is the **non-directional change in allele frequencies** caused by random transmission in a population of finite size





Key Concepts

Population Genetics

Our genomes contain a historical record of human evolution

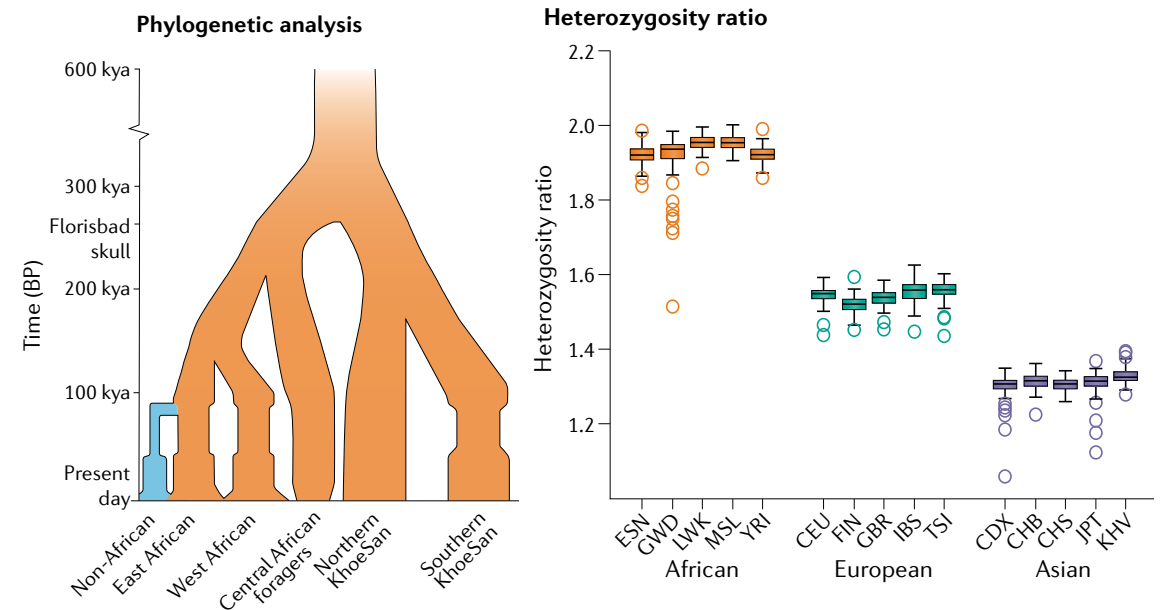
- Unique features of genetic architecture in African populations
 - Most genetically diverse
 - Higher levels of heterozygosity increase the discoverability of new variants
 - Smaller LD blocks of correlated variants

African genetic diversity and adaptation inform a precision medicine agenda

[Luisa Pereira](#), [Leon Mutesa](#), [Paulina Tindana](#) & [Michèle Ramsay](#)

[Nature Reviews Genetics](#) 22, 284–306 (2021) | [Cite this article](#)

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Key Concepts

Population Genetics

Selection causes directional shifts in allele frequency

- **Selective sweep:** a strongly advantageous allele is driven from low to high frequency

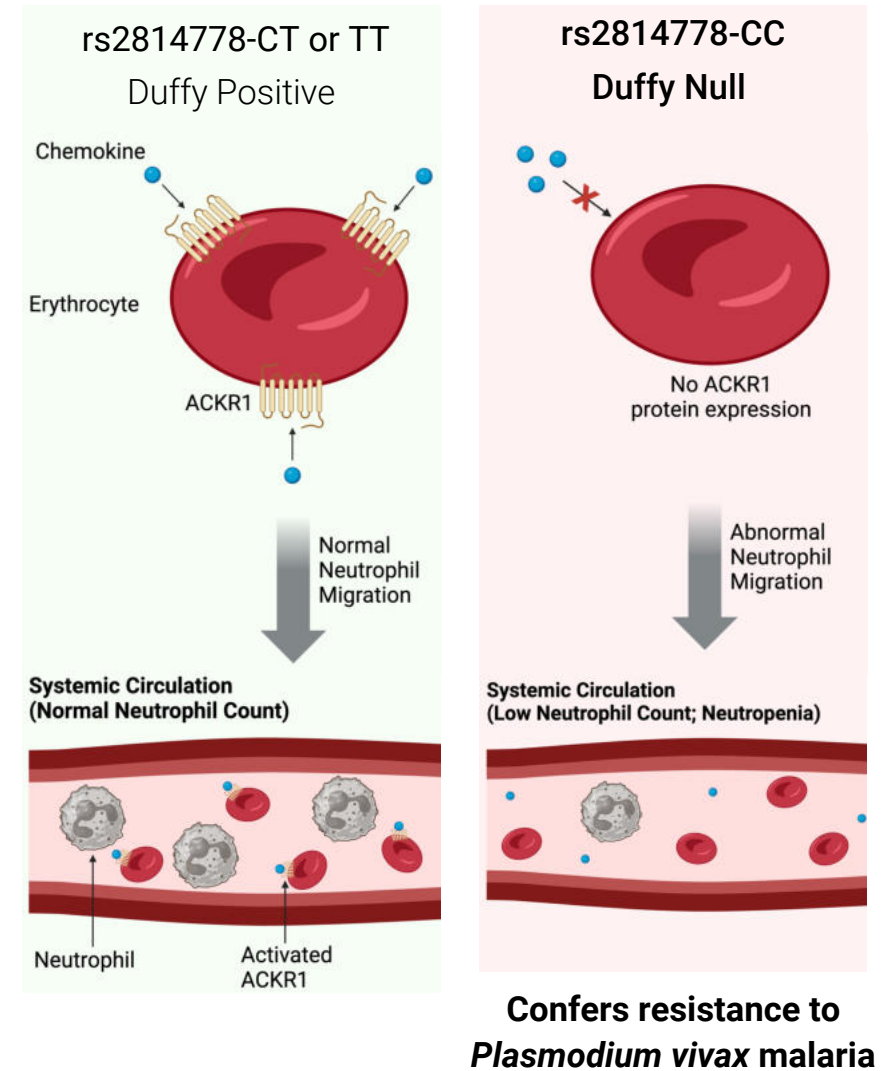


Key Concepts

Population Genetics

Selection causes directional shifts in allele frequency

- **Selective sweep:** a strongly advantageous allele is driven from low to high frequency
- **Soft sweep:** mutation drifts initially, but becomes favored after an environmental change
 - rs2814778-CC:
 - 90% in Western sub-Saharan Africa
 - 50% in the Arabian Peninsula
 - 5% to 20% across India
 - 0.2% in Europe



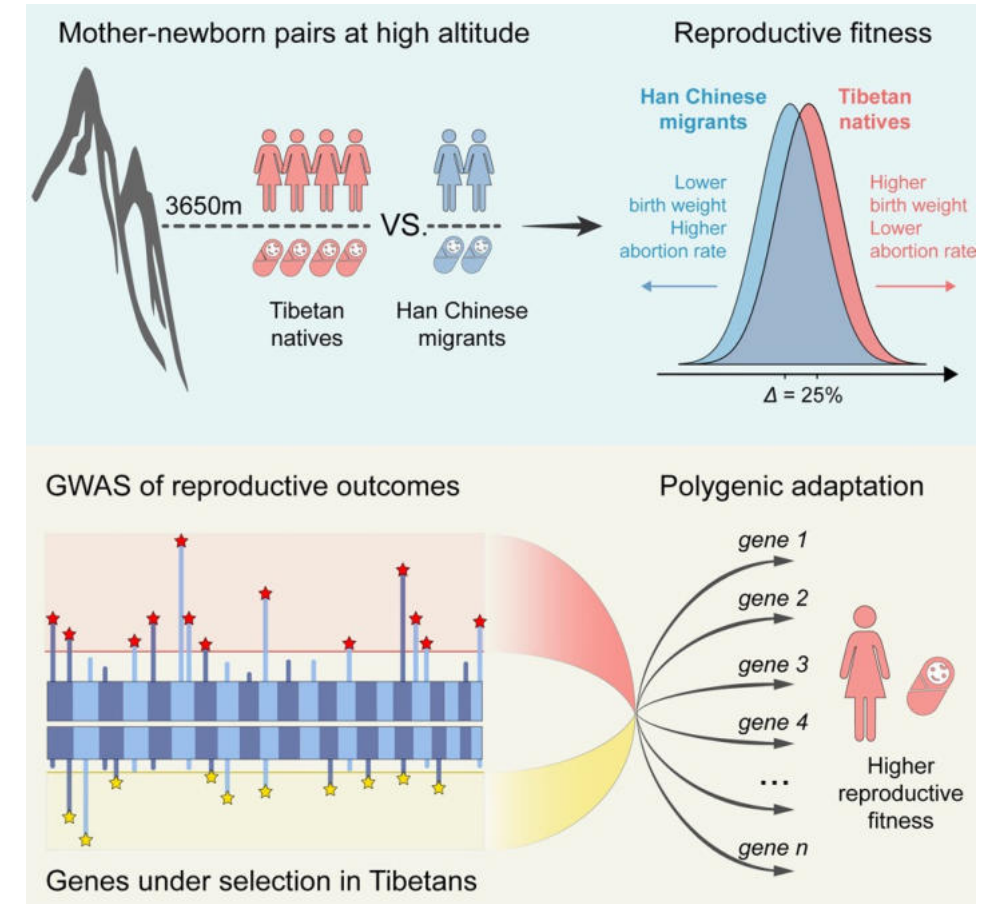


Key Concepts

Population Genetics

Selection causes directional shifts in allele frequency

- **Selective sweep:** a strongly advantageous allele is driven from low to high frequency
- **Soft sweep:** mutation drifts initially, but becomes favored after an environmental change
- **Polygenic adaptation:** small shifts in allele frequencies spread across many loci
 - Most relevant for complex phenotypes, including diseases like cancer



He et al. *Current Biology* (2023)



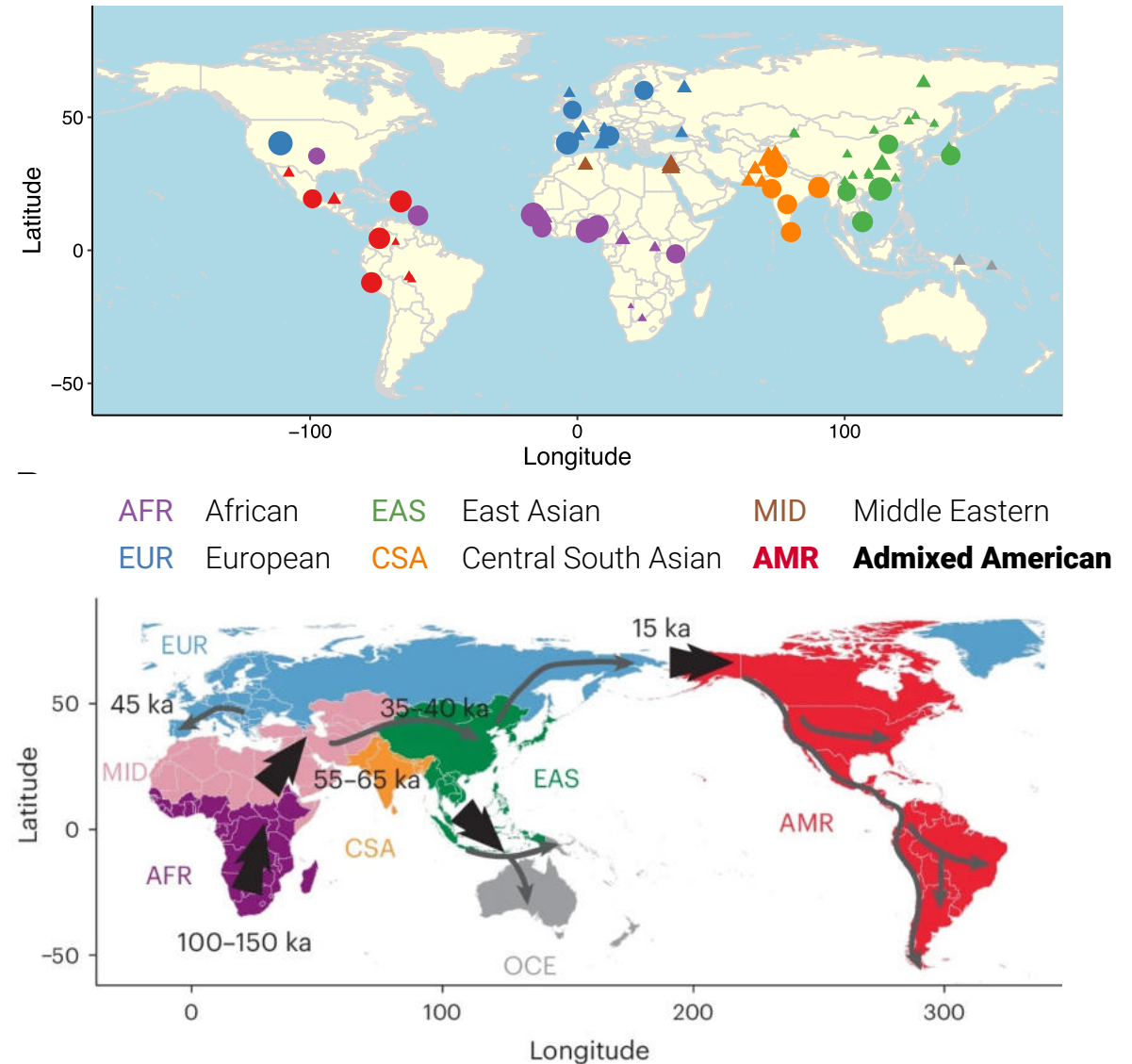
Key Concepts

Population Structure

Genetic ancestry refers to stretches of DNA that we inherit from our ancestors

- In most studies, **genetic ancestry is a statistical estimate of genetic similarity with respect to a reference population**
 - Often at the continental level
- “Admixed” individuals have recent ancestry from more than population, but all modern humans have multiple ancestries

1000 Genomes Project (1KGP)
Human Genome Diversity Project (HGDP)



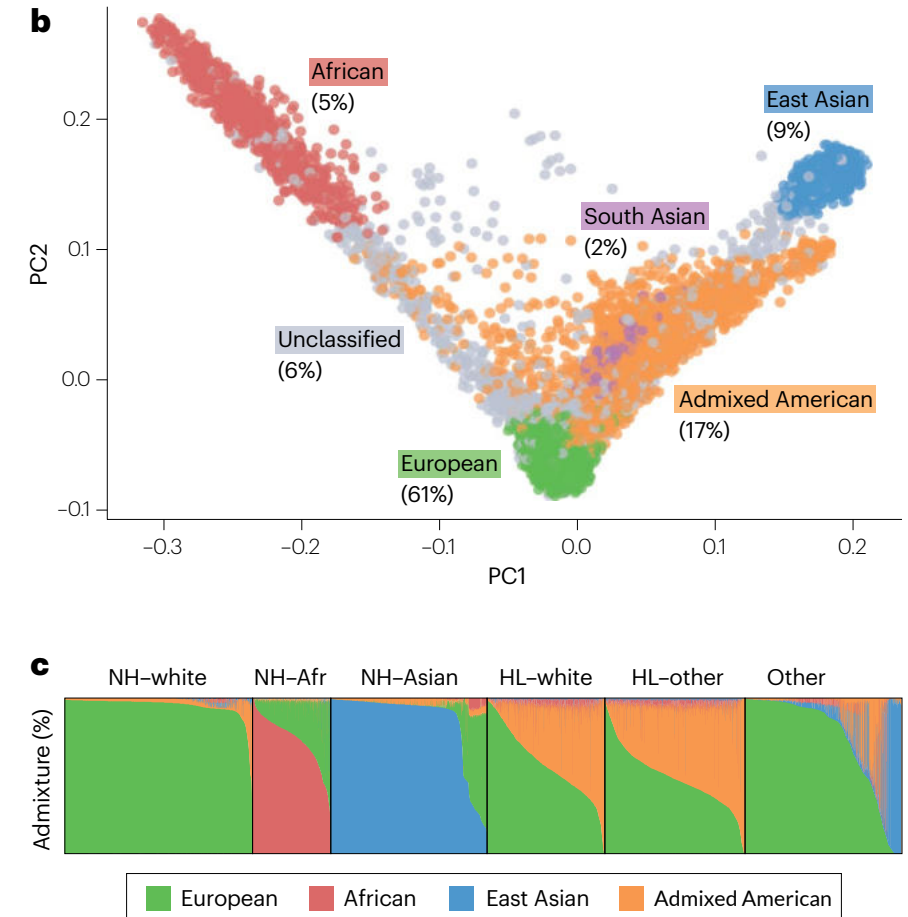


Key Concepts

Population Structure

Global genetic ancestry: an individual's overall, genome-wide ancestry composition profile

- Clustering-based methods:
 - Principal component analysis (PCA)
 - STRUCTURE/ADMIXTURE estimate proportions attributed to each ancestry
 - Many individuals can be unclassified and excluded from analysis
- Should not be conflated with race ethnicity



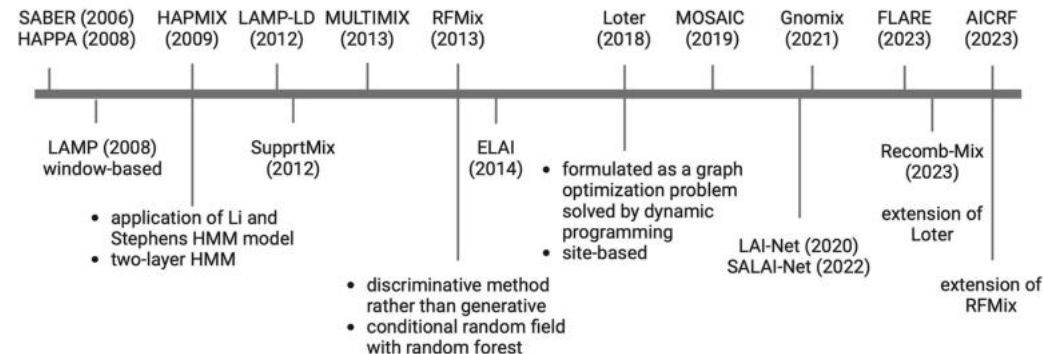
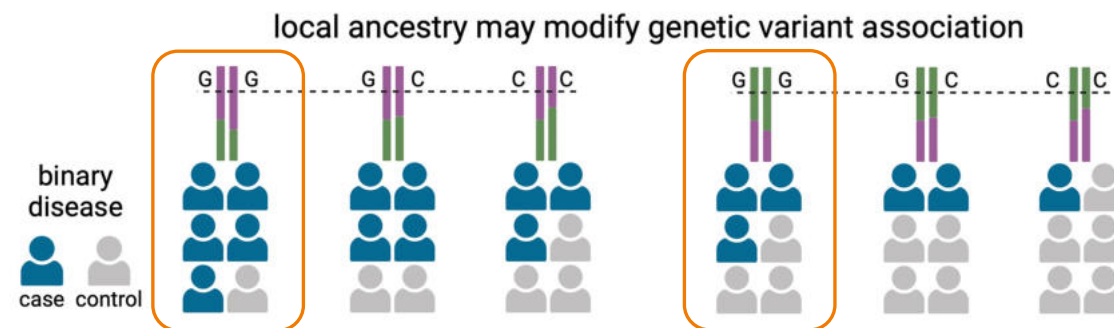
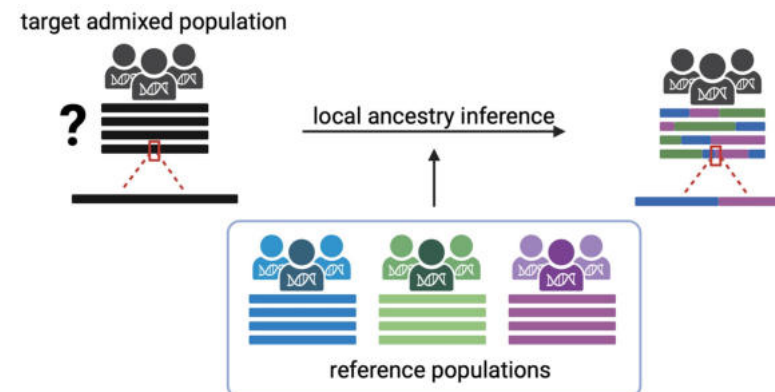


Key Concepts

Population Structure

Local genetic ancestry: number of alleles inherited from a given ancestry at a specific locus or position in the genome

- Methods use hidden Markov models (HMM) and **machine learning algorithms** like random forest and support vector machine
- High resolution view of population structure





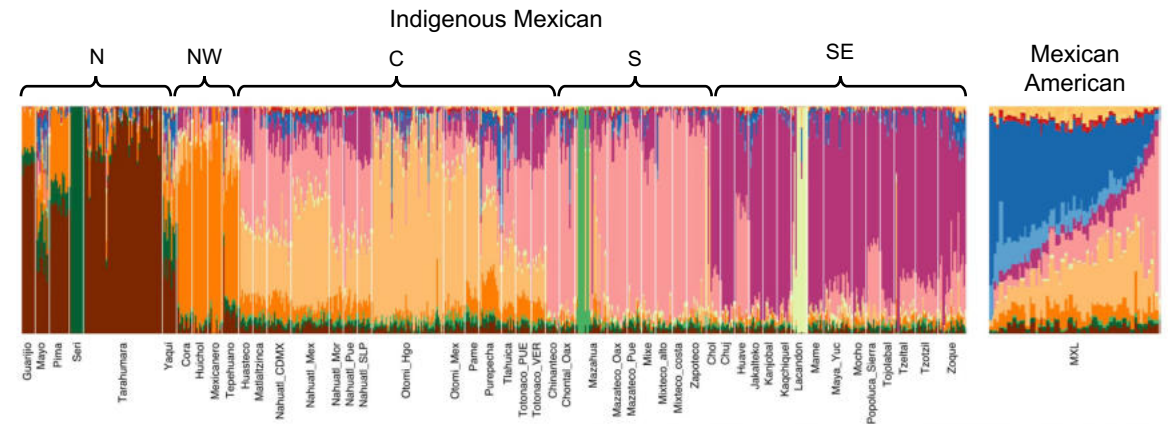
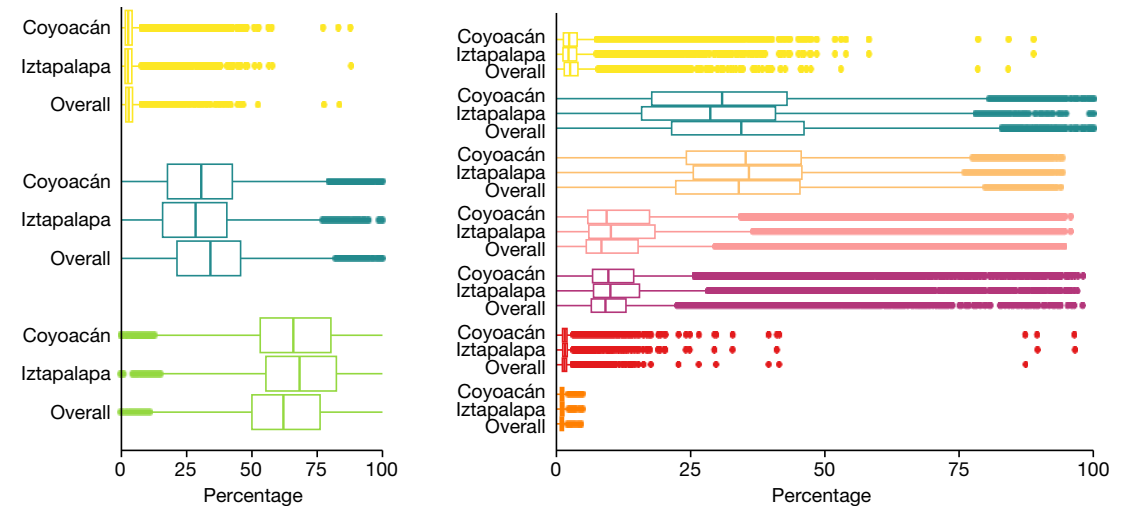
Local genetic ancestry: number of alleles inherited from a given ancestry at a specific locus or position in the genome

- **High resolution view of population structure**
 - Can be used to derive more accurate genome-wide ancestry estimates
 - **Resolution depends on reference panel**
(eg: advantage of using Indigenous Mexican reference for MCPS)

■ African
 ■ Indigenous Mexican
 ■ African
 ■ South Mexican

■ European
 ■ European
 ■ Southeast Mexican

■ Central Mexican
 ■ North Mexican
 ■ Northwest Mexican



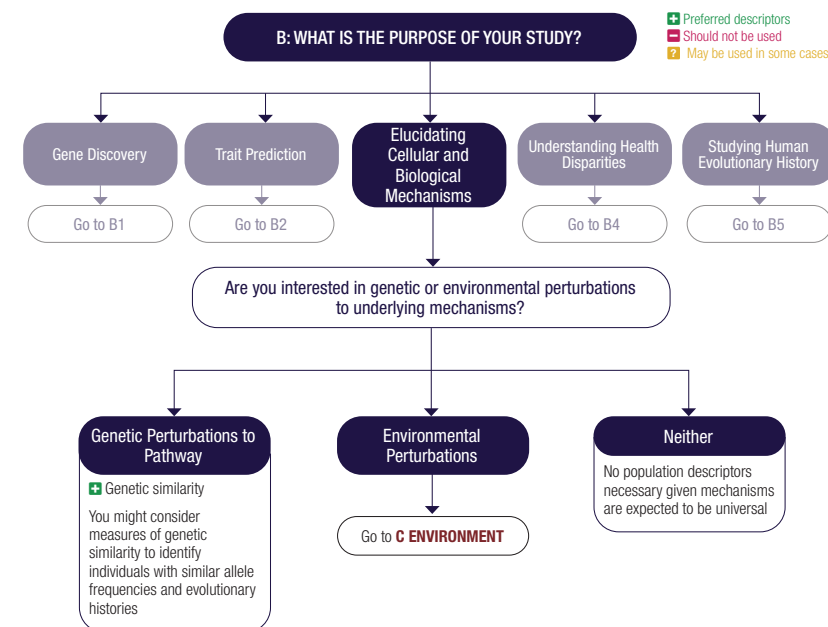


Key Concepts

Population Groupings

How we **group and label populations** matters for research validity, clinical utility, and health equity

- Human genetic variation exists on a continuum
 - **Avoid typological thinking**
 - Carefully describe and justify groupings
- Genetic ancestry is not an “objective” measure
 - Estimated genetic similarity depends on data inputs (eg: reference panel) and methods
 - Reference populations are not homogeneous



Recommendations for responsible use of population descriptors in polygenic risk score development

[Johanna L. Smith](#), [Clement A. Adebamowo](#), [Sally N. Adebamowo](#), [Burcu F. Darst](#), [Stephanie M. Fullerton](#), [Stephanie M. Gogarten](#), [Marwan E. Hamed](#), [Jibril B. Hirbo](#), [Micah R. Hysong](#), [Angad Singh Johar](#), [Alyna T. Khan](#), [Iftikhar J. Kullo](#), [Iain R. Konigsberg](#), [Peter Kraft](#), [Leslie A. Lange](#), [Yun Li](#), [Alicia R. Martin](#), [Sarah C. Nelson](#), [Ananyo Choudhury](#), [Michèle Ramsay](#), [Ewan K. Cobran](#), [Daniel J. Schaid](#), [Jayati Sharma](#), [Ying Wang](#), [Polygenic Risk Methods Development \(PRIMED\) Consortium](#) & ... [Quan Sun](#)

[Nature Genetics](#) 57, 2962–2971 (2025)



Part II:

Leveraging population diversity to study cancer



Introduction

GWAS

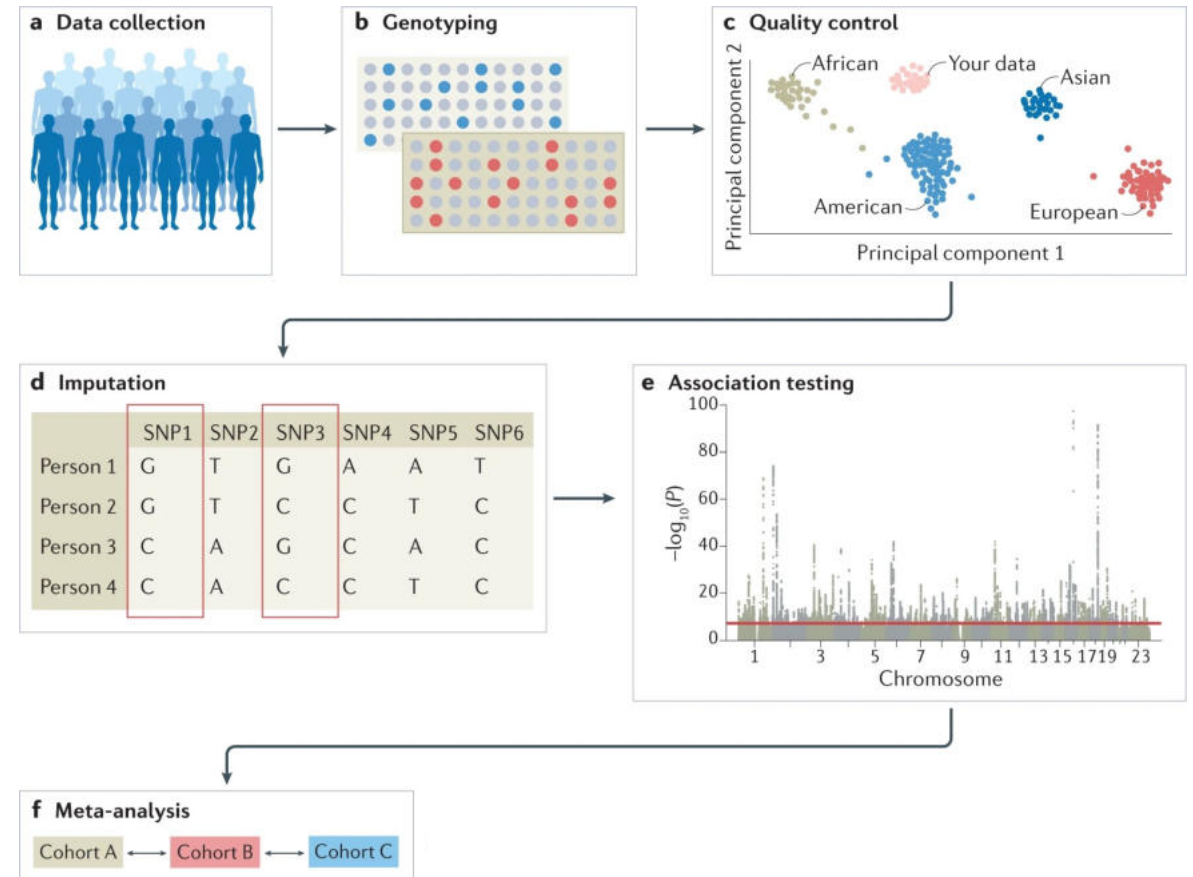
Genome-wide association studies (GWAS)

- Case-control studies of unrelated individuals
 - Mostly sporadic cancer cases
- Hypothesis-free scan of millions of common ($MAF \geq 0.01$) variants across the genome
 - Associations with **p-value $< 5 \times 10^{-8}$** are considered statistically significant, lower p-values do not imply causality
- Increase power by performing meta-analysis

Genome-wide association studies

[Emil Uffelmann](#), [Qin Qin Huang](#), [Nchangwi Syntia Munung](#), [Jantina de Vries](#), [Yukinori Okada](#), [Alicia R. Martin](#), [Hilary C. Martin](#), [Tuuli Lappalainen](#) & [Danielle Posthuma](#) ✉

[Nature Reviews Methods Primers](#) **1**, Article number: 59 (2021) | [Cite this article](#)



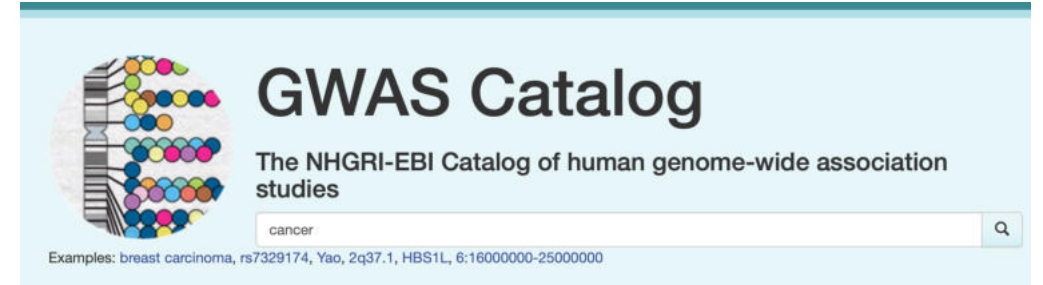


Cancer GWAS

Overview

What can GWAS tell us about cancer etiology?

- Characterize genetic architecture
- Strengthen causal inference for risk factors with a genetic basis
- Decipher biological mechanisms
 - Integrate with multi-omic (eg: metQTL, pQTL) and functional data (eg: ATAC-seq, MPRA, CRISPR)

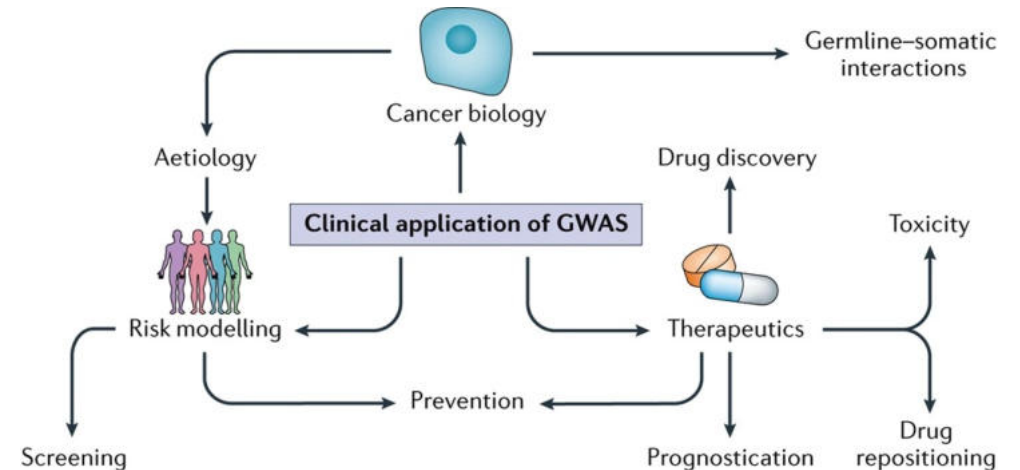


Search results for *cancer*

T cancer MONDO_0004992

A tumor composed of atypical neoplastic, often pleomorphic cells that invade other tissues. Malignant neoplasms often metastasize to distant anatomic sites and may recur after excision. The most commo... [Show more >](#)

Associations **18902** Studies **1798**



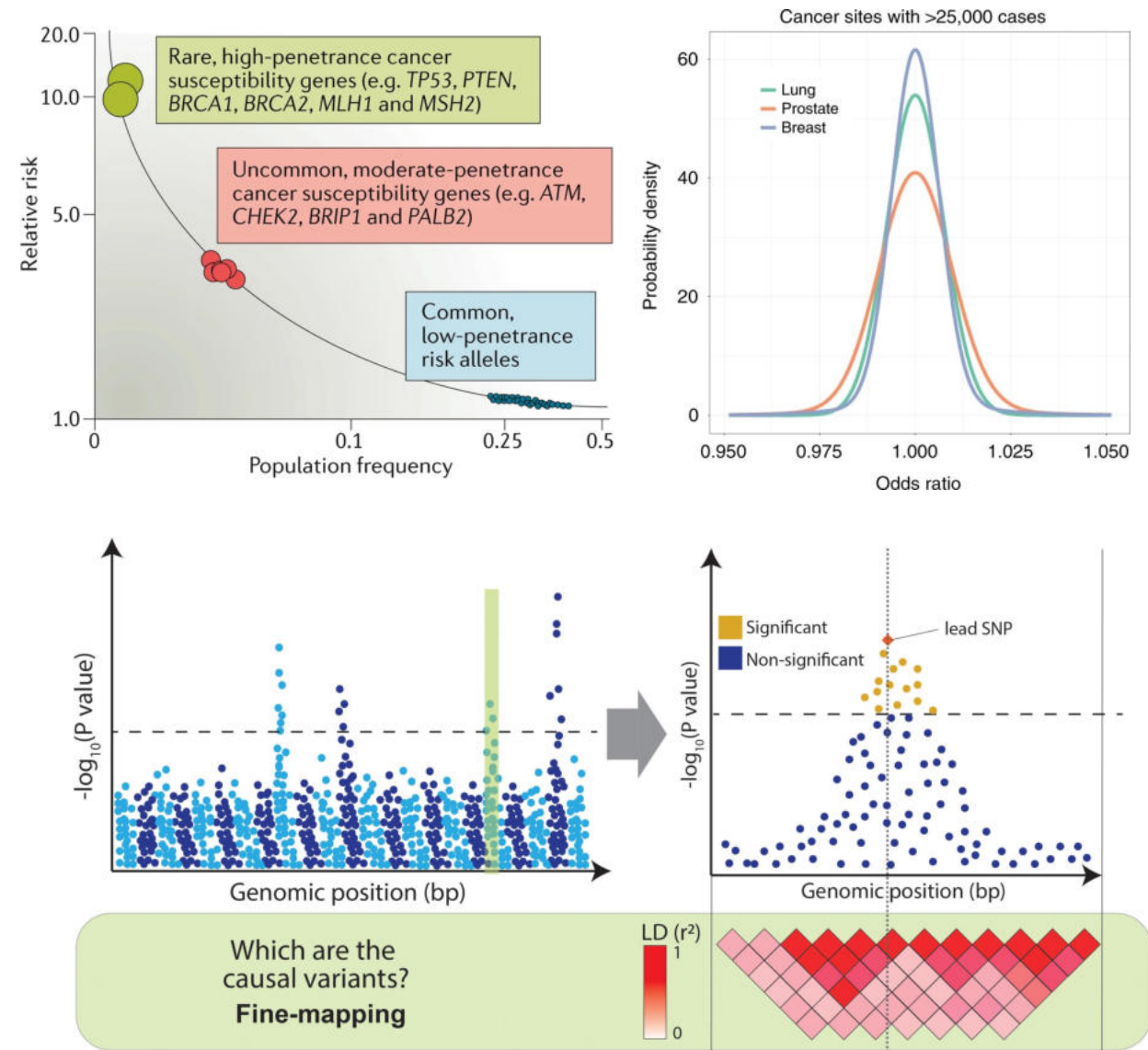


Cancer GWAS

Genetic Architecture

What can GWAS tell us about cancer etiology?

- Characterize genetic architecture
 - Common variant effect sizes are small
 - Distribution of effect sizes across risk loci (**sparsity vs. polygenicity**)
 - Uncover “causal” variants using statistical **fine-mapping**





Cancer GWAS

Genetic Architecture

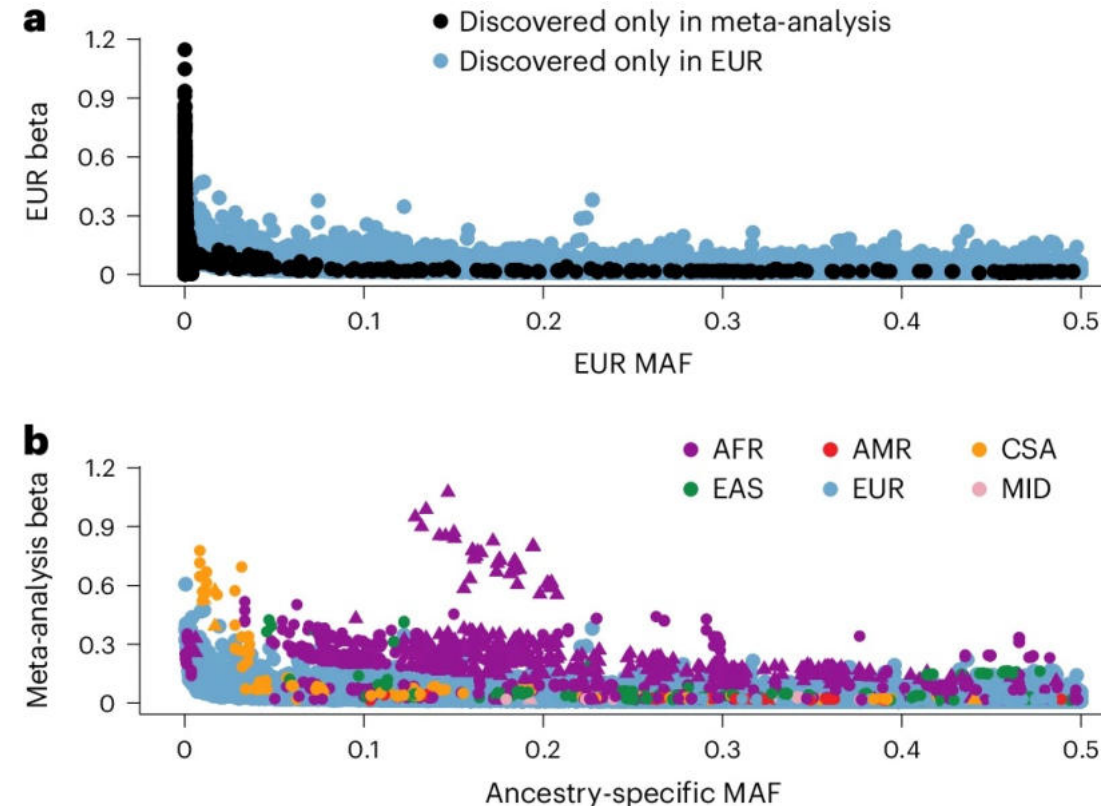
Population diversity improves **power for discovery** and **detection of causal variants**

- Wider spectrum of allelic variation increases power for discovery
- Novel loci discovered by including non-EUR ancestries in meta-analysis:
 - Million Veteran Program: 3,477 loci (13.3%) non-EUR 39% of total sample size
 - Pan-UK Biobank: 14,676 loci (6.2%) non-EUR 4.5% of total sample size

Pan-UK Biobank genome-wide association analyses enhance discovery and resolution of ancestry-enriched effects

[Konrad J. Karczewski](#), [Rahul Gupta](#), [Masahiro Kanai](#), [Wenhan Lu](#), [Kristin Tsuo](#), [Ying Wang](#), [Raymond K. Walters](#), [Patrick Turley](#), [Shawneequa Callier](#), [Nirav N. Shah](#), [Nikolas Bawa](#), [Duncan S. Palmer](#), [Jacqueline I. Goldstein](#), [Gopal Sarma](#), [Matthew Solomonson](#), [Nathan Cheng](#), [Sam Bryant](#), [Claire Churchhouse](#), [Caroline M. Cusick](#), [Timothy Poterba](#), [John Compitello](#), [Daniel King](#), [Wei Zhou](#), [Cotton Seed](#), ... [Alicia R. Martin](#)

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Cancer GWAS

Genetic Architecture

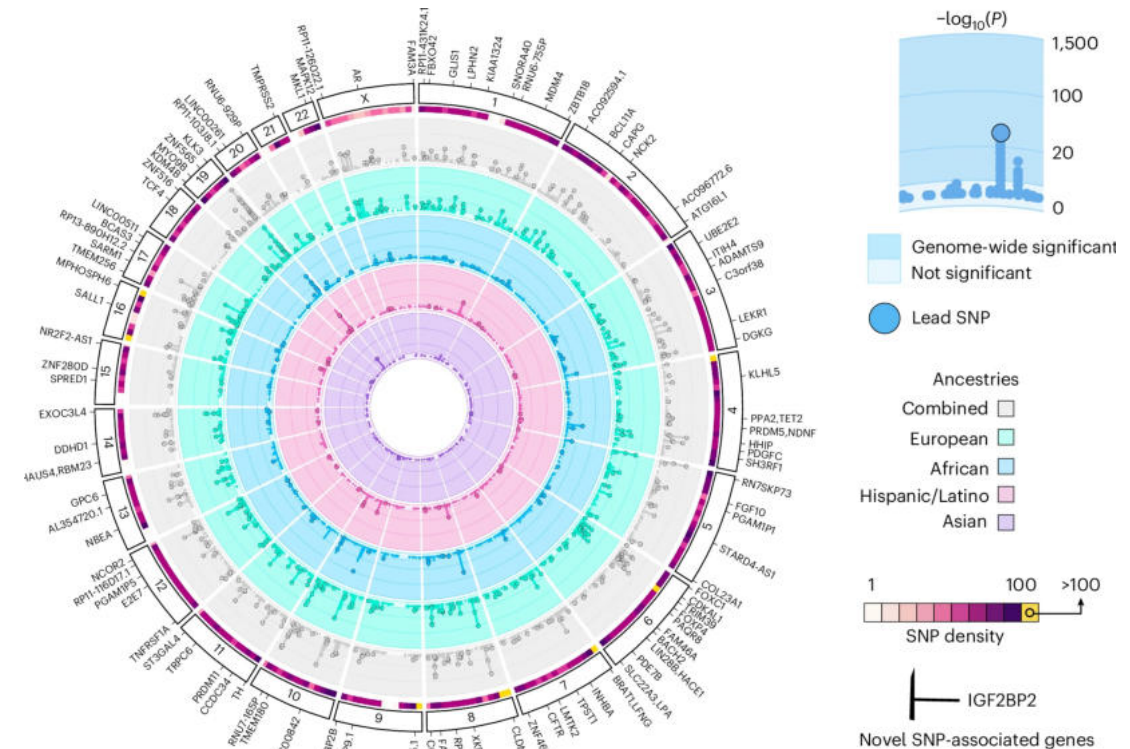
Population diversity improves **power for discovery** and **detection of causal variants**

- Prostate-specific antigen (PSA) is a biomarker that can be used to screen for prostate cancer
 - Million Veteran Program: N=296,754 non-EUR **28%** of total sample size
 - 184** novel independent signals ($p < 5 \times 10^{-8}$)
 - Novel loci discovered by including non-EUR ancestries in meta-analysis: **72 (39%)**

Genome-wide association study of prostate-specific antigen levels in 392,522 men identifies new loci and improves prediction across ancestry groups

[Thomas J. Hoffmann](#), [Rebecca E. Graff](#), [Ravi K. Madduri](#), [Alex A. Rodriguez](#), [Clinton L. Cario](#), [Karen Feng](#), [Yu Jiang](#), [Anqi Wang](#), [Robert J. Klein](#), [Brandon L. Pierce](#), [Scott Eggner](#), [Lin Tong](#), [William Blot](#), [Jirong Long](#), [Louisa B. Goss](#), [Burcu F. Darst](#), [Timothy Rebbeck](#), [Joseph Lachance](#), [Caroline Andrews](#), [Akindele O. Adebisi](#), [Ben Adusei](#), [Oseremen I. Aisuodionoe-Shadrach](#), [Pedro W. Fernandez](#), [Mohamed Jalloh](#), ... [Linda Kachuri](#) & [John S. Witte](#)

Nature Genetics 57, 334–344 (2025)





Cancer GWAS

Genetic Architecture

Population diversity improves **power for discovery** and **detection of causal variants**

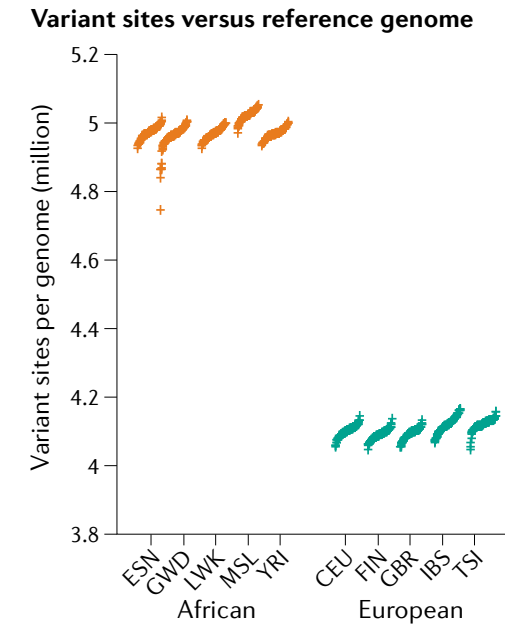
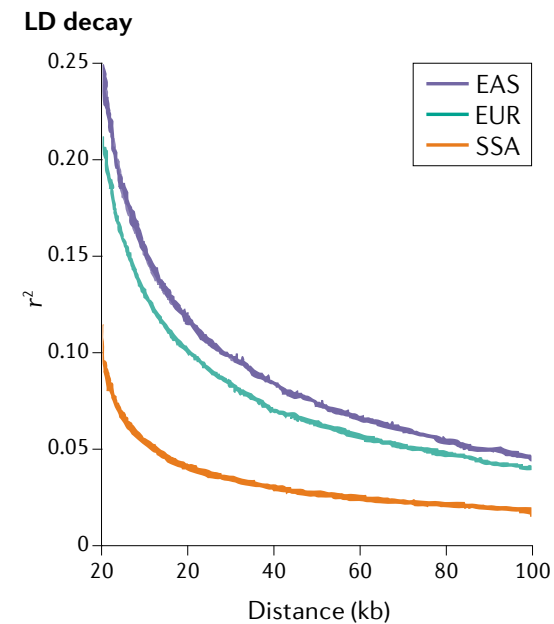
- Improved resolution for identifying causal variants using fine-mapping
 - More variants and shorter LD blocks in AFR/AFR descent populations reduce the number of variants correlated with causal variant(s)

African genetic diversity and adaptation inform a precision medicine agenda

[Luisa Pereira](#), [Leon Mutesa](#), [Paulina Tindana](#) & [Michèle Ramsay](#)

[Nature Reviews Genetics](#) **22**, 284–306 (2021) | [Cite this article](#)

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Cancer GWAS

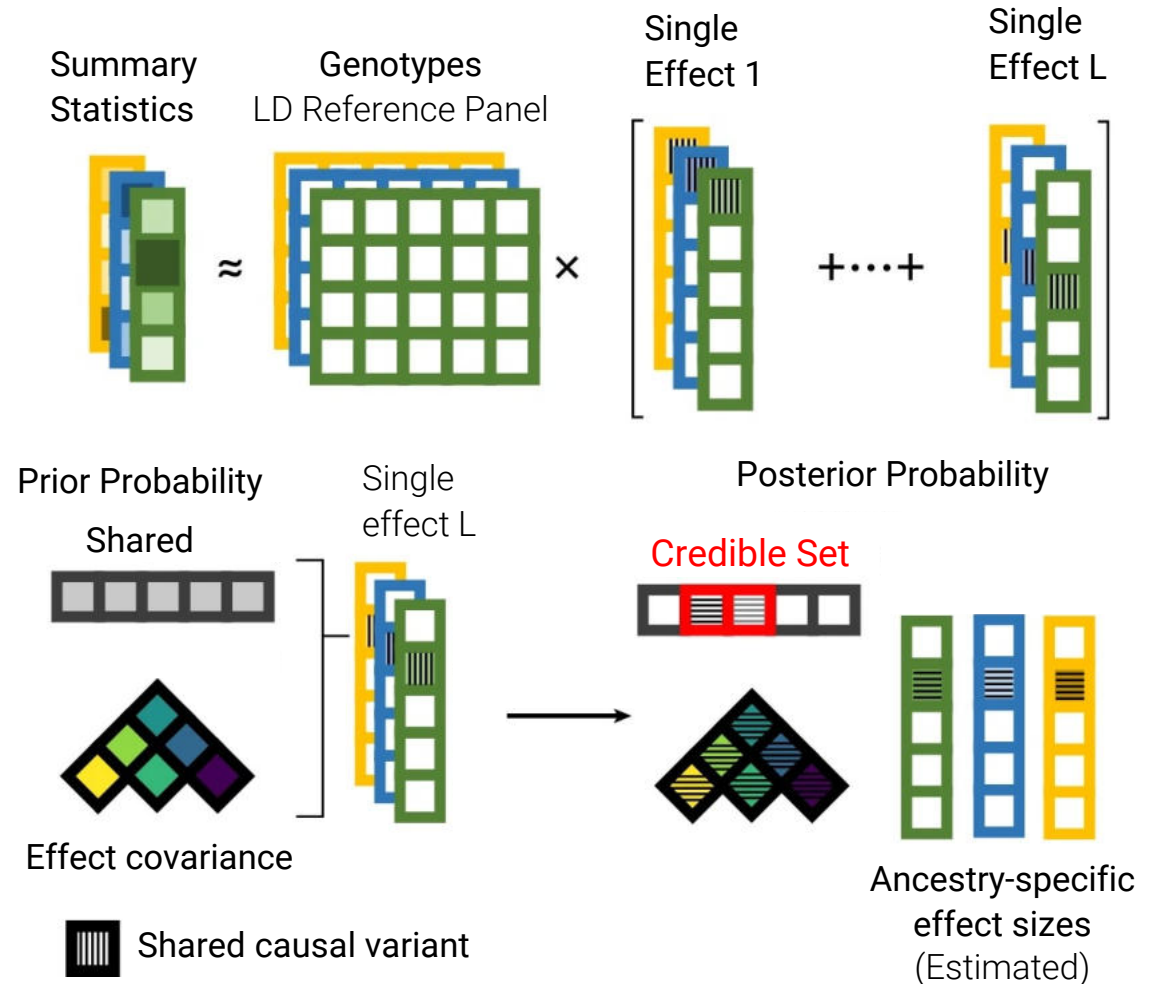
Genetic Architecture

Population diversity improves **power for discovery** and **detection of causal variants**

- Improved resolution for identifying causal variants using fine-mapping
 - More variants and shorter LD blocks in AFR/AFR descent populations reduce the number of variants correlated with causal variant(s)
 - **Smaller 95% credible sets:** total number of likely causal variants with **PIPs (posterior inclusion probability)** that add up to 95%

Improved multiancestry fine-mapping identifies *cis*-regulatory variants underlying molecular traits and disease risk

Zeyun Lu , Xinran Wang, Matthew Carr, Artem Kim, Steven Gazal, Pejman Mohammadi, Lang Wu, James Pirruccello, Linda Kachuri, Alexander Gusev & Nicholas Mancuso
Nature Genetics 57, 1881–1889 (2025)





Cancer GWAS

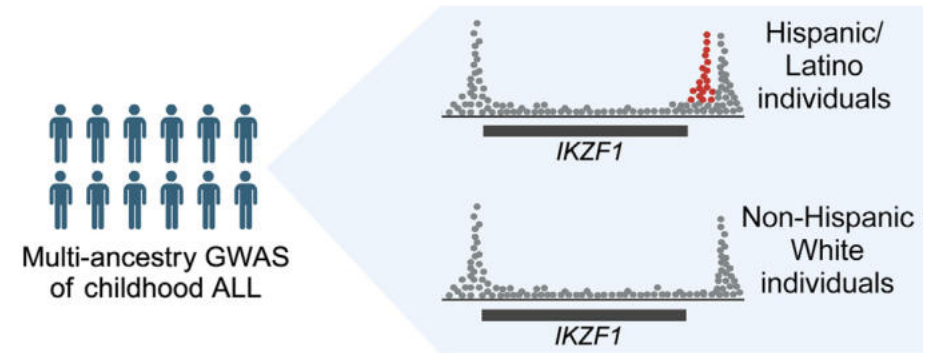
Genetic Architecture

Population diversity improves **power for discovery** and **detection of causal variants**

- Childhood acute lymphoblastic leukemia (ALL)
 - Incidence is significantly higher in Latino children in the US and in Central America

A noncoding regulatory variant in *IKZF1* increases acute lymphoblastic leukemia risk in Hispanic/Latino children

Adam J. de Smith^{1 2 13}  , Lara Wahlster^{3 4 13}, Soyoung Jeon^{1 2 13}, Linda Kachuri⁵, Susan Black^{3 4}, Jalen Langie^{1 2}, Liam D. Cato^{3 4}, Nathan Nakatsuka⁶, Tsz-Fung Chan^{1 2}, Guangze Xia⁷, Soumyaa Mazumder^{3 4}, Wenjian Yang⁸, Steven Gazal^{1 2}, Celeste Eng^{9 10}, Donglei Hu⁹, Esteban González Burchard^{9 10}, Elad Ziv⁹, Catherine Metayer¹¹, Nicholas Mancuso^{1 2}, Jun J. Yang⁸...Vijay G. Sankaran^{3 4 14 15}  





Cancer GWAS

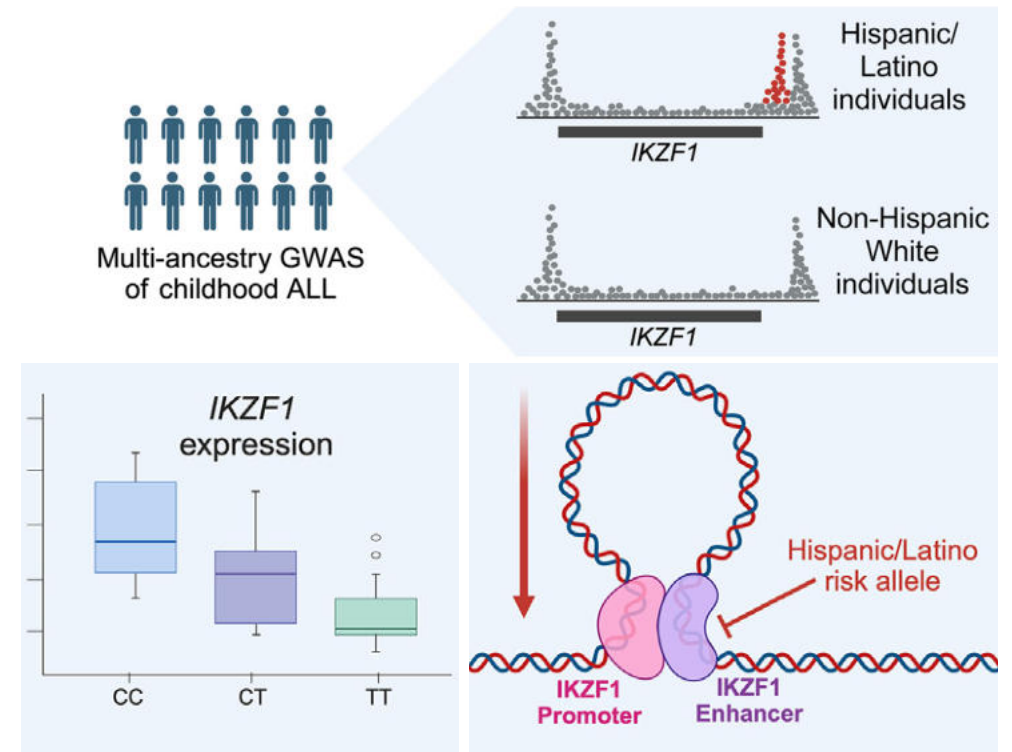
Genetic Architecture

Population diversity improves **power for discovery** and **detection of causal variants**

- New independent ALL signal in *IKZF1* found in Latino children from California
 - rs76880433 (OR=1.44) tags functional causal variant rs1451367
 - Disrupts *IKZF1* enhancer in B-cell progenitors
 - Reduces *IKZF1* expression in Mexican American children

A noncoding regulatory variant in *IKZF1* increases acute lymphoblastic leukemia risk in Hispanic/Latino children

Adam J. de Smith^{1 2 13}  , Lara Wahlster^{3 4 13}, Soyoung Jeon^{1 2 13}, Linda Kachuri⁵, Susan Black^{3 4}, Jalen Langie^{1 2}, Liam D. Cato^{3 4}, Nathan Nakatsuka⁶, Tsz-Fung Chan^{1 2}, Guangze Xia⁷, Soumyaa Mazumder^{3 4}, Wenjian Yang⁸, Steven Gazal^{1 2}, Celeste Eng^{9 10}, Donglei Hu⁹, Esteban González Burchard^{9 10}, Elad Ziv⁹, Catherine Metayer¹¹, Nicholas Mancuso^{1 2}, Jun J. Yang⁸...Vijay G. Sankaran^{3 4 14 15}  





Cancer GWAS

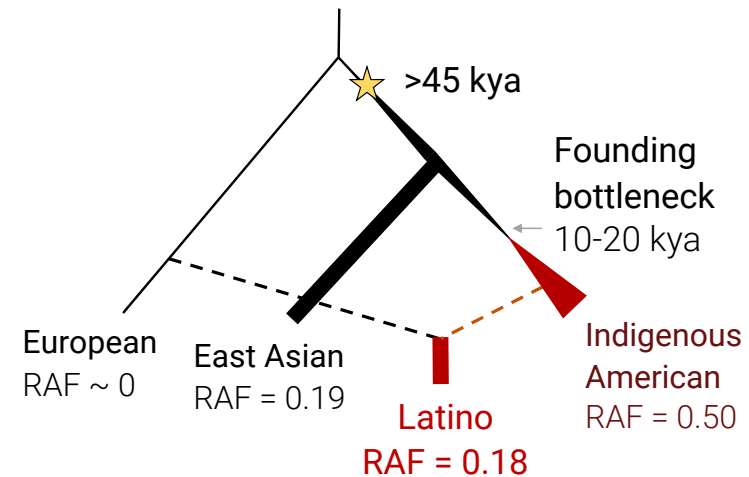
Genetic Architecture

Population diversity improves **power for discovery** and **detection of causal variants**

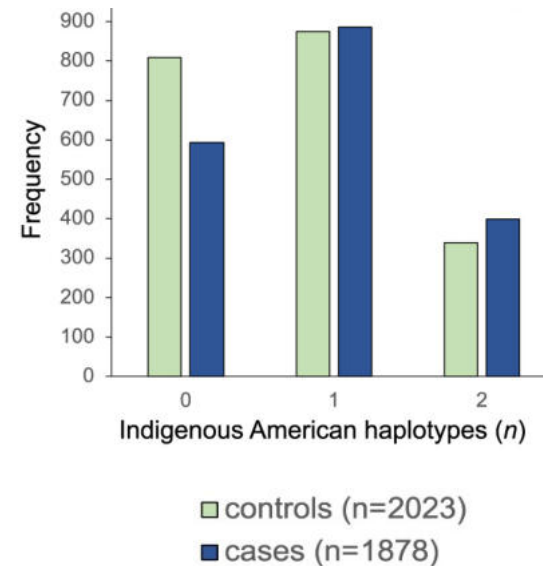
- New independent ALL signal in *IKZF1* found in Latino children from California
- Risk allele frequency (RAF) reflects genealogy
- Risk haplotype enriched in cases ($p=8.8 \times 10^{-7}$) and under positive selection in Mexican and Peruvian populations
 - rs1451367 present in oldest sequenced Indigenous American (12,700 years old)

A noncoding regulatory variant in *IKZF1* increases acute lymphoblastic leukemia risk in Hispanic/Latino children

Adam J. de Smith^{1 2 13}  , Lara Wahlster^{3 4 13}, Soyoung Jeon^{1 2 13}, Linda Kachuri⁵, Susan Black^{3 4}, Jalen Langie^{1 2}, Liam D. Cato^{3 4}, Nathan Nakatsuka⁶, Tsz-Fung Chan^{1 2}, Guangze Xia⁷, Soumya Mazumder^{3 4}, Wenjian Yang⁸, Steven Gazal^{1 2}, Celeste Eng^{9 10}, Donglei Hu⁹, Esteban González Burchard^{9 10}, Elad Ziv⁹, Catherine Metayer¹¹, Nicholas Mancuso^{1 2}, Jun J. Yang⁸...Vijay G. Sankaran^{3 4 14 15}  



Reconstructed genealogies
using 1000 Genomes samples



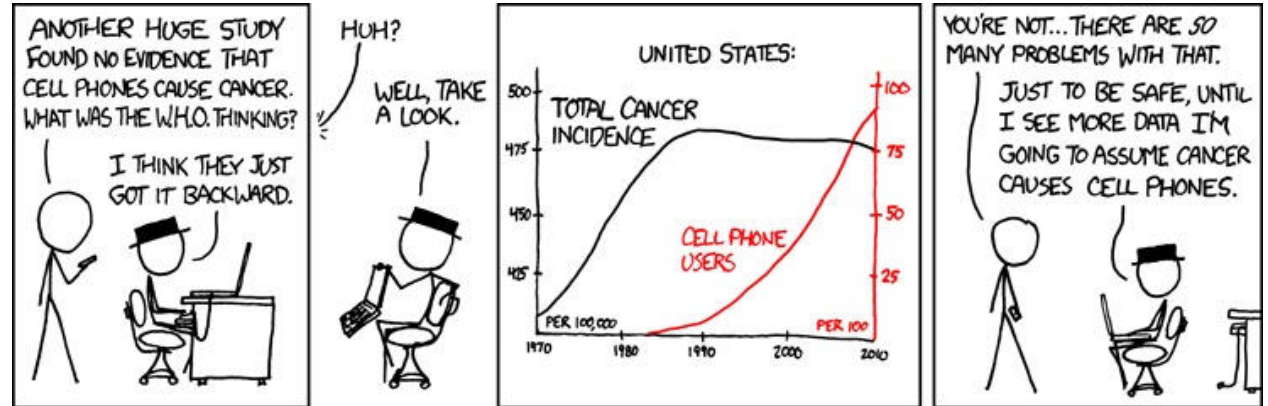


Cancer GWAS

Causality

What can GWAS tell us about cancer etiology?

- Strengthen causal inference for risk factors with a genetic basis: **Mendelian Randomization (MR)**



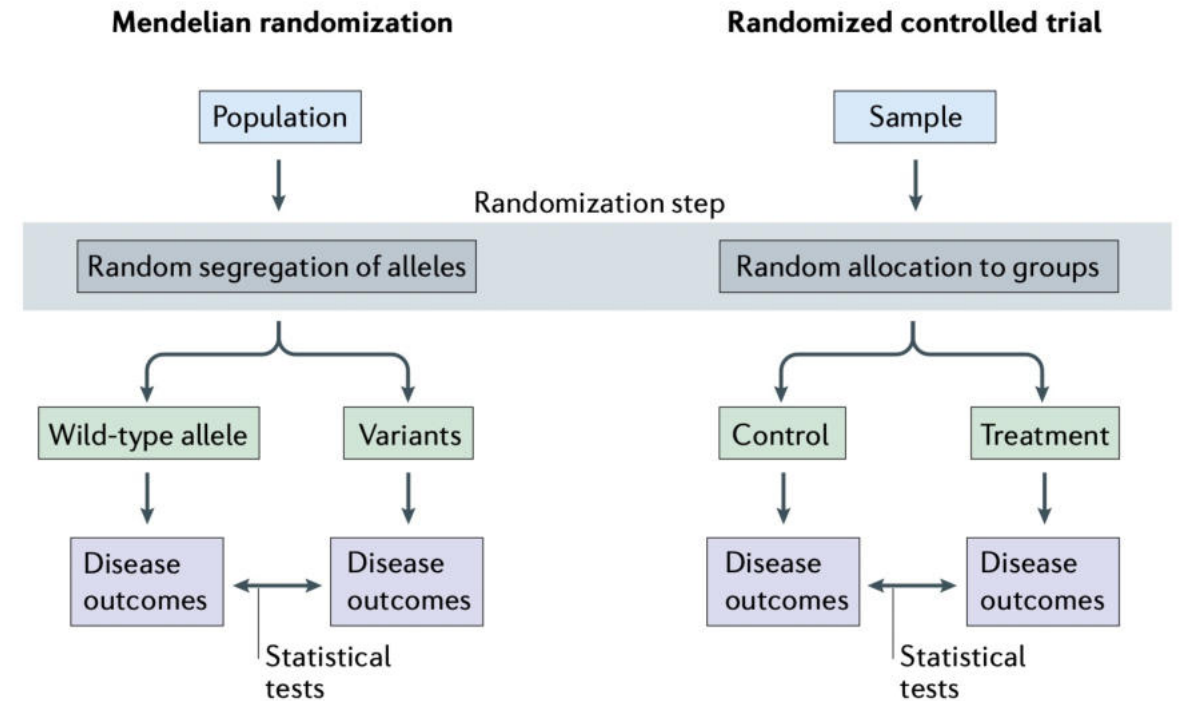


Cancer GWAS

Causality

What can GWAS tell us about cancer etiology?

- Strengthen causal inference for risk factors with a genetic basis:
Mendelian Randomization (MR)
 - Genetic variants used as instrumental variables to infer causal relationships
 - Overcomes reverse causation and confounding in observational studies
- Not all exposures are amenable to MR
 - Avoid social, behavioral phenotypes



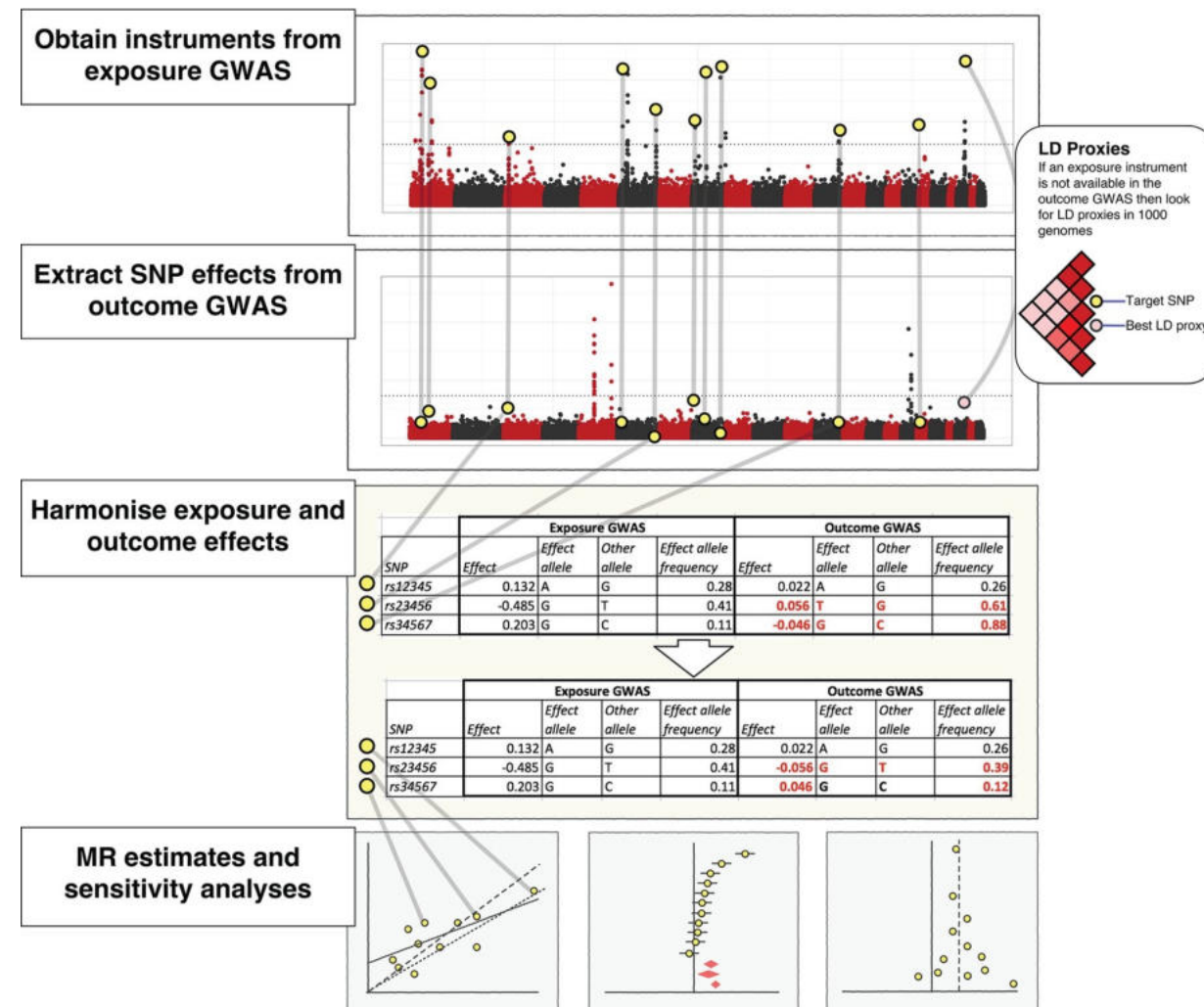


Cancer GWAS

Causality

What can GWAS tell us about cancer etiology?

- Strengthen causal inference for cancer risk factors with a genetic basis:
Mendelian Randomization (MR)
 - Genetic variants used as instrumental variables to infer causal relationships
 - Exposure-outcome relationship is tested by using marginal genetic effects on the exposure and outcome





Cancer GWAS

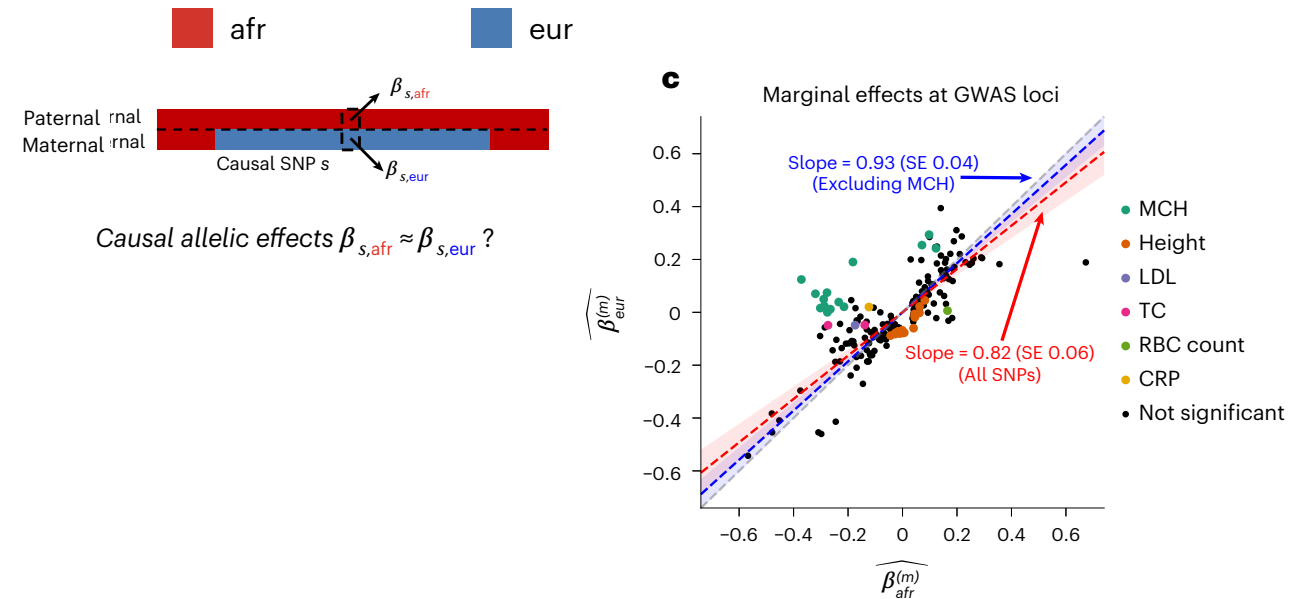
Causality

Multi-ancestry perspective on causal inference

Causal effects are conserved across ancestries

- Valid instruments should replicate MR conclusions across populations

Causal effects on complex traits are similar for common variants across segments of different continental ancestries within admixed individuals





Cancer GWAS

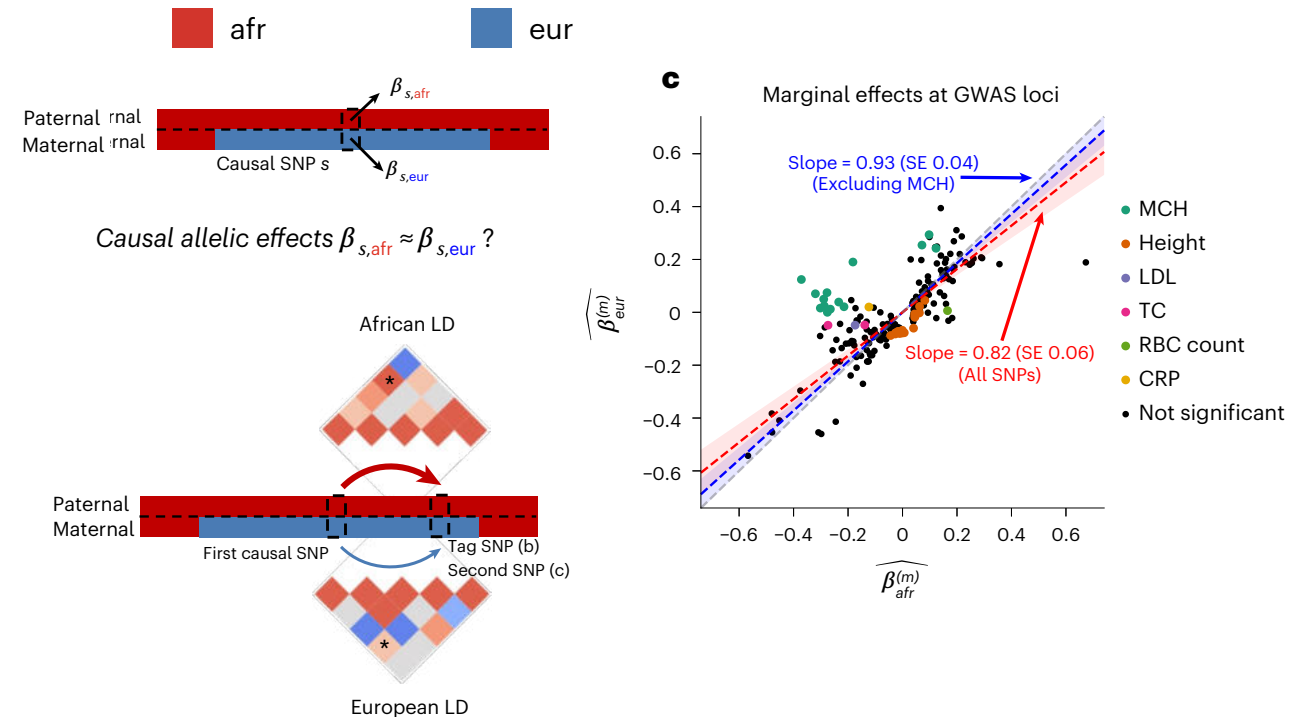
Causality

Multi-ancestry perspective on causal inference

Causal effects are conserved across ancestries

- Valid instruments should replicate MR conclusions across populations
 - Marginal GWAS effects for non-causal (tagging) variants can be heterogeneous due to LD and local ancestry
 - Marginal effects for functional variants (eg: eQTL) are highly consistent

Causal effects on complex traits are similar for common variants across segments of different continental ancestries within admixed individuals



Hou et al. *Nature Genetics* (2023)

Kachuri et al. *Nature Genetics* (2023), Taylor et al. *Nature* (2024)



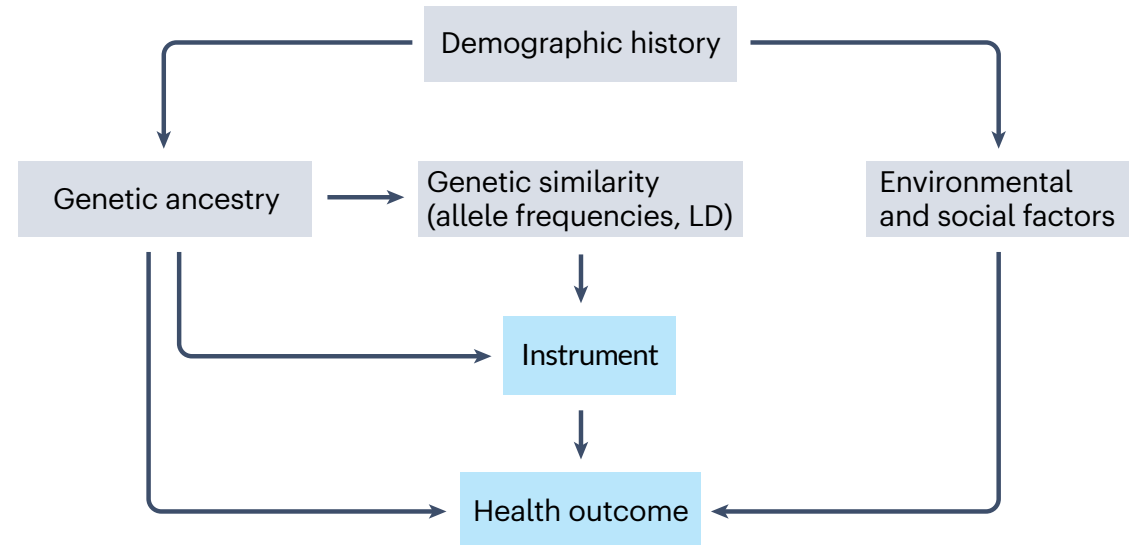
Cancer GWAS

Causality

Multi-ancestry perspective on causal inference

Leverage unique population characteristics

- Effect of folate in populations without supplementation
- Effect of alcohol in East Asian populations based on functional *ALDH2* variant with frequency = 30-50%
 - **Caution:** rs671 increases cancer risk by affecting alcohol *metabolism* not consumption



Different confounding structures can make MR replication and interpretation challenging



Beyond GWAS

Somatic Landscape

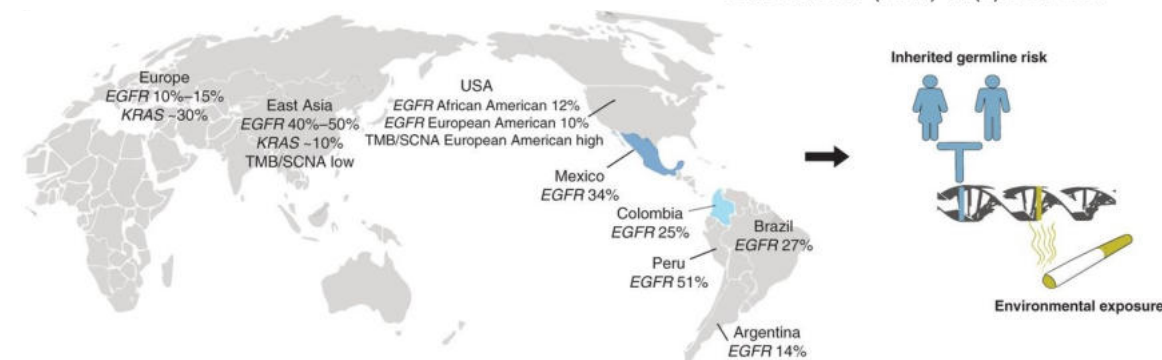
How does genetic diversity impact studies of somatic features?

- Differences in the prevalence of driver mutations across populations
 - Can reflect differences in exposure

Genetic Ancestry Contributes to Somatic Mutations in Lung Cancers from Admixed Latin American Populations

Jian Carrot-Zhang; Giovanni Soca-Chafre ; Nick Patterson; Aaron R. Thorner ; Anwesha Nag; Jacqueline Watson; Giulio Genovese; July Rodriguez; Maya K. Gelbard; Luis Corrales-Rodriguez; Yoichiro Mitsuishi; Gavin Ha ; Joshua D. Campbell ; Geoffrey R. Oxnard; Oscar Arrieta ; Andres F. Cardona ; Alexander Gusev ; Matthew Meyerson

Cancer Discov (2021) 11 (3): 591–598.



EGFR alteration prevalence				
EUR	AFR	EAS	SAS	AMR
14.6%	15.7%	53.4%	36.4%	30.3%

KRAS alteration prevalence				
EUR	AFR	EAS	SAS	AMR
38.9%	32.5%	15.0%	20.5%	23.1%

Meta-analysis reveals differences in somatic alterations by genetic ancestry across common cancers

Setor Amuzu, Amy X. Xie, Xuechun Bai, Kelly R. Pekala, Nicholas A. Pickersgill, David Ma, Tomin Perea-Chamblee, Kanika Arora, Walid K. Chatila, Andriy Derkach, Ronglai Shen, Garrett M. Frampton, Michael Berger, Nikolaus Schultz, Justin Y. Newberg & Jian Carrot-Zhang

Nature Genetics 57, 2655–2660 (2025) | [Cite this article](#)



Beyond GWAS

Somatic Landscape

How does genetic diversity impact studies of somatic features?

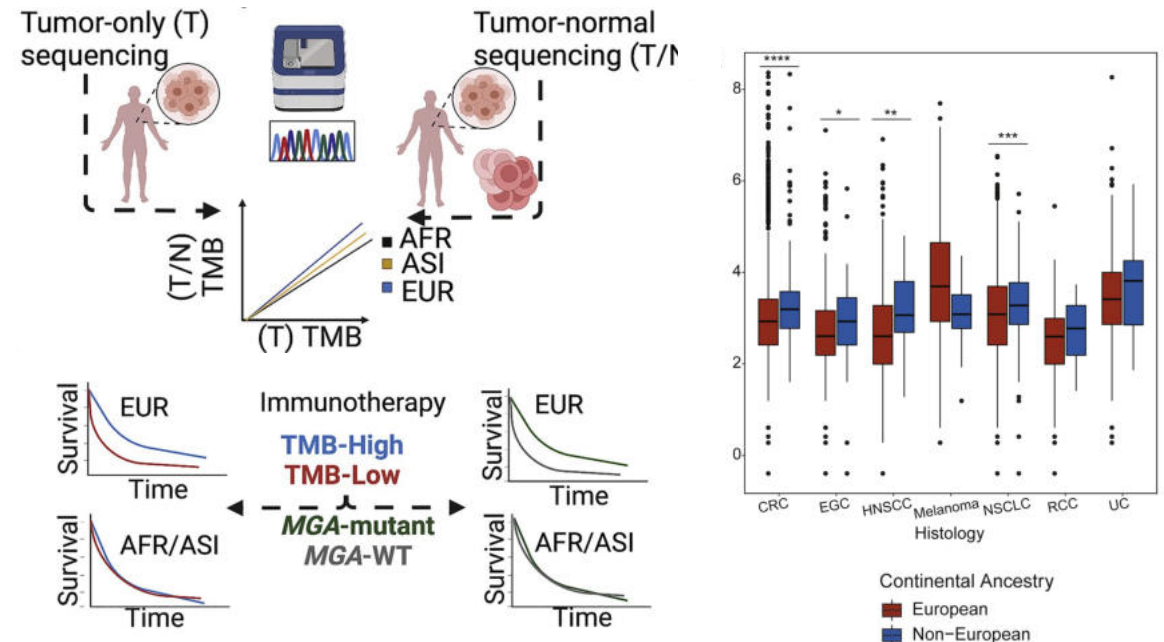
- Differences in the prevalence of driver mutations across populations
 - Can reflect differences in exposure
- Differences in the distribution of tumor mutational burden (TMB)
 - May reflect misclassification due to contamination in tumor-only panels

Collect germline data ☺

Cancer Cell

Ancestry-driven recalibration of tumor mutational burden and disparate clinical outcomes in response to immune checkpoint inhibitors

Amin H. Nassar^{1 2 3 22}, Elio Adib^{2 3 22}, Sarah Abou Alaiwi^{2 3 22}, Talal El Zarif², Stefan Groha^{4 5}, Elie W. Akl³, Pier Vitale Nuzzo², Tarek H. Mouhieddine⁶, Tomin Perea-Chamblee⁷, Kodi Taraszka⁸, Habib El-Khoury^{9 10}, Muhieddine Labban¹¹, Christopher Fong⁷, Kanika S. Arora^{12 13}, Chris Labaki², Wenxin Xu², Guru Sonpavde², Robert I. Haddad¹⁰, Kent W. Mouw¹⁴, Marios Giannakis¹⁰...Alexander Gusev^{4 5 20 21 23}  





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